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**Nakayama**

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(54) **SENSOR APPARATUS**

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(52) **U.S. Cl.**

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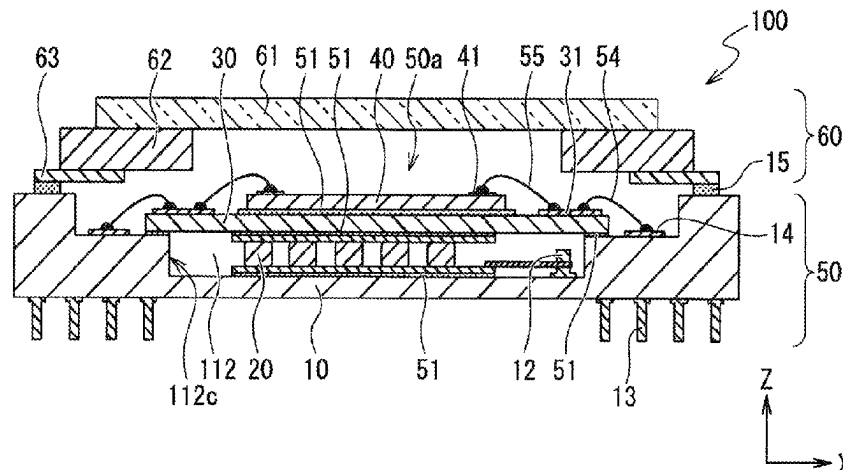
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(57)

**ABSTRACT**

Provided is a sensor apparatus that can suppress movement of foreign matter from a Peltier element to a sensor element. The sensor apparatus includes a package substrate, a Peltier element, a circuit substrate, and a sensor element. The package substrate has a recess portion on a side of a first surface and plural terminals on a side of a second surface located on an opposite side of the first surface. The Peltier element is arranged in the recess portion. The circuit substrate is arranged on an opposite side of a bottom surface of the recess portion with the Peltier element sandwiched

(Continued)



therebetween. The sensor element is attached to an opposite side of a surface, the surface being opposed to the Peltier element.

# 10 Claims, 13 Drawing Sheets

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**H10F 77/60** (2025.01)

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 27/14636; H01L 23/13; G01J 1/0271;  
 G01J 1/44; G01J 2001/448

See application file for complete search history.

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FIG. 1

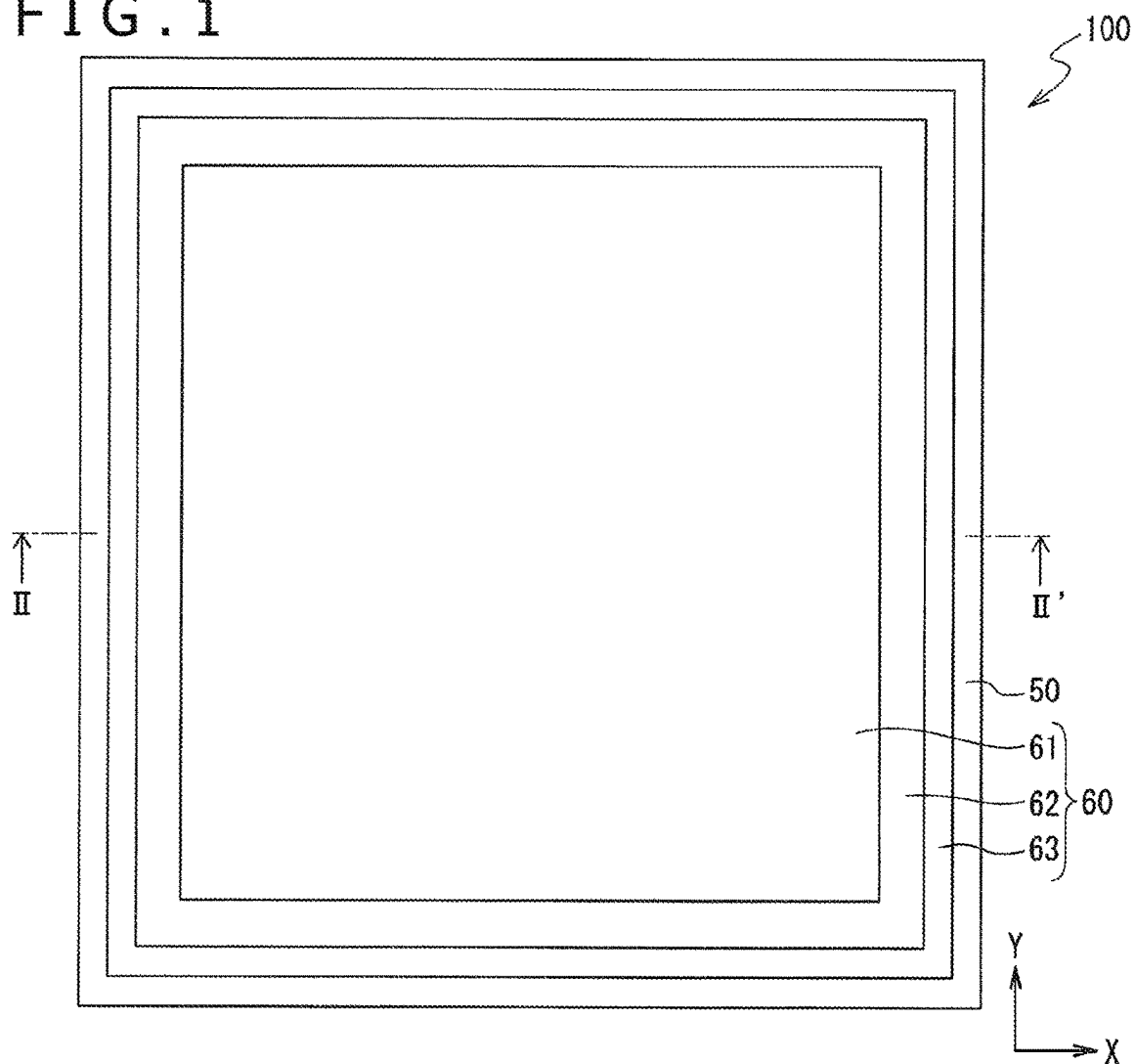


FIG. 2

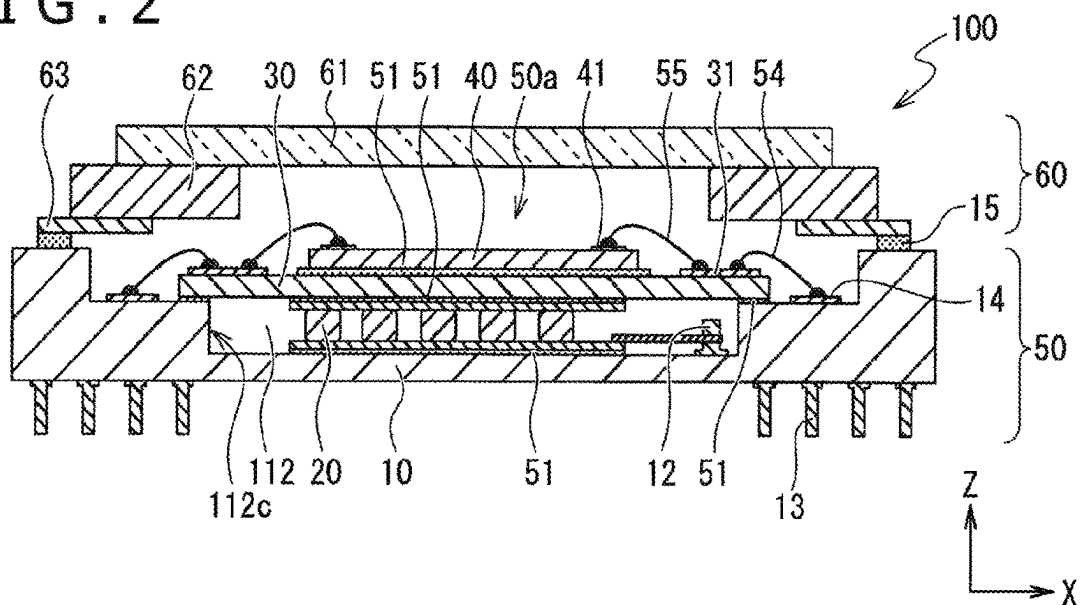


FIG. 3

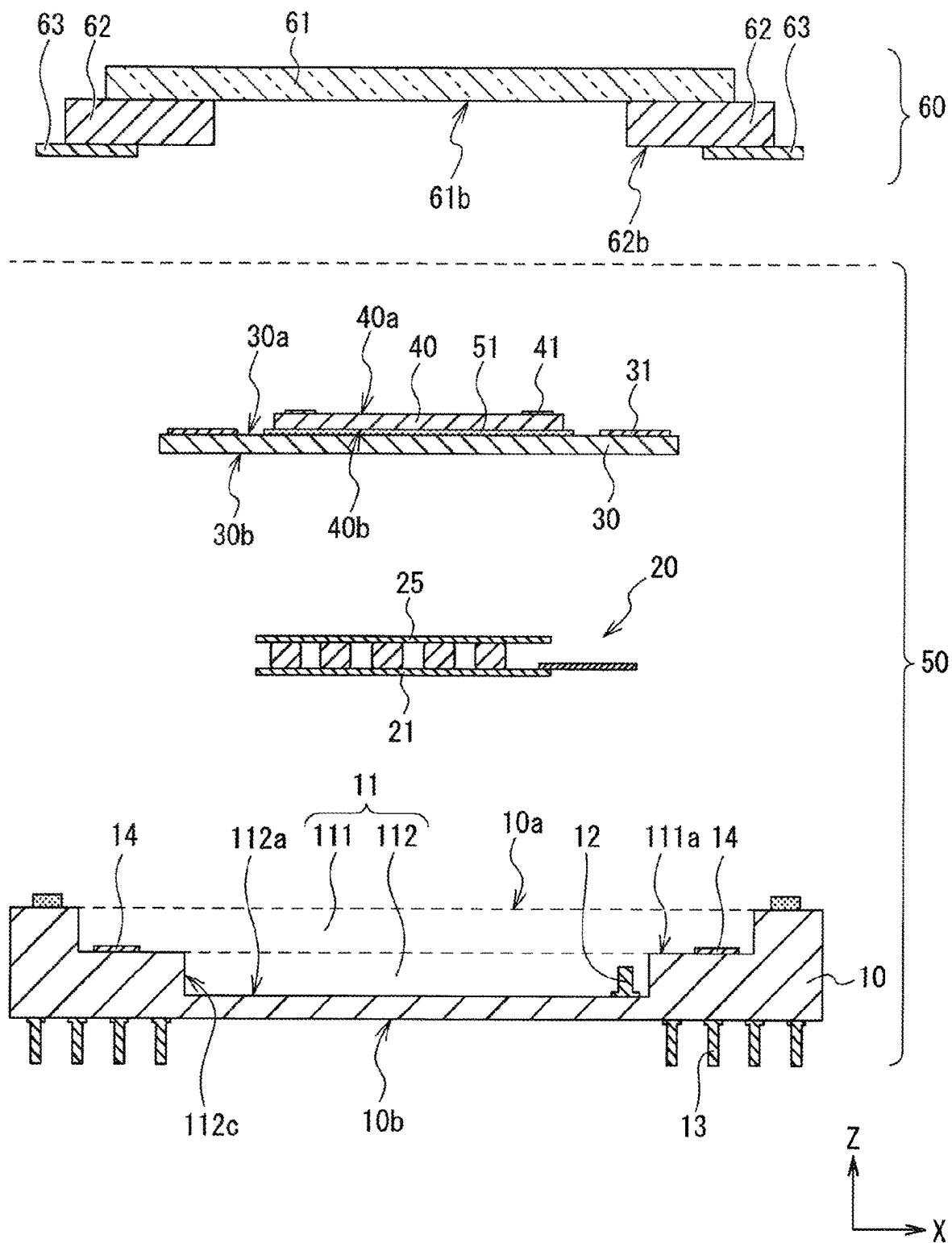


FIG. 4

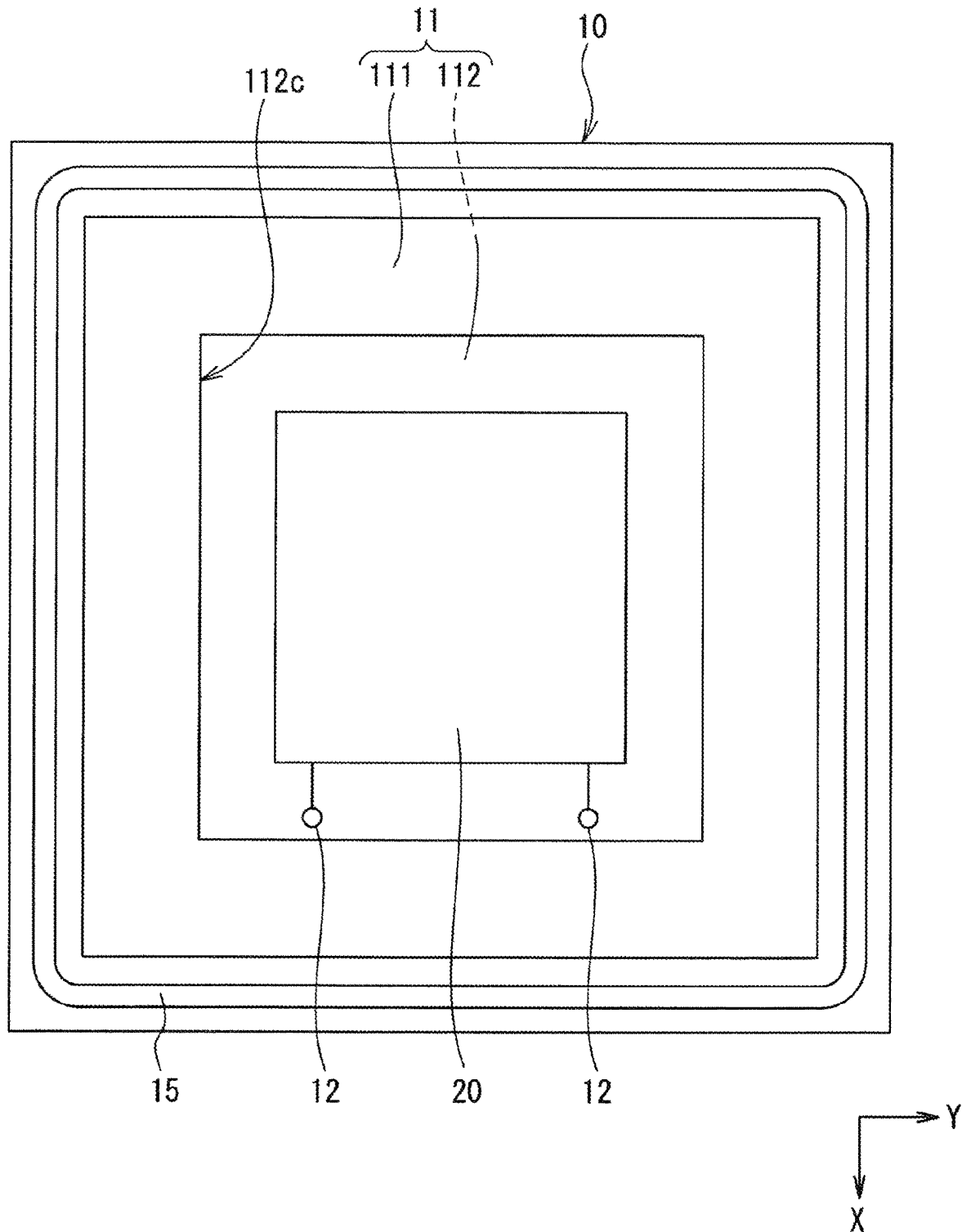


FIG. 5

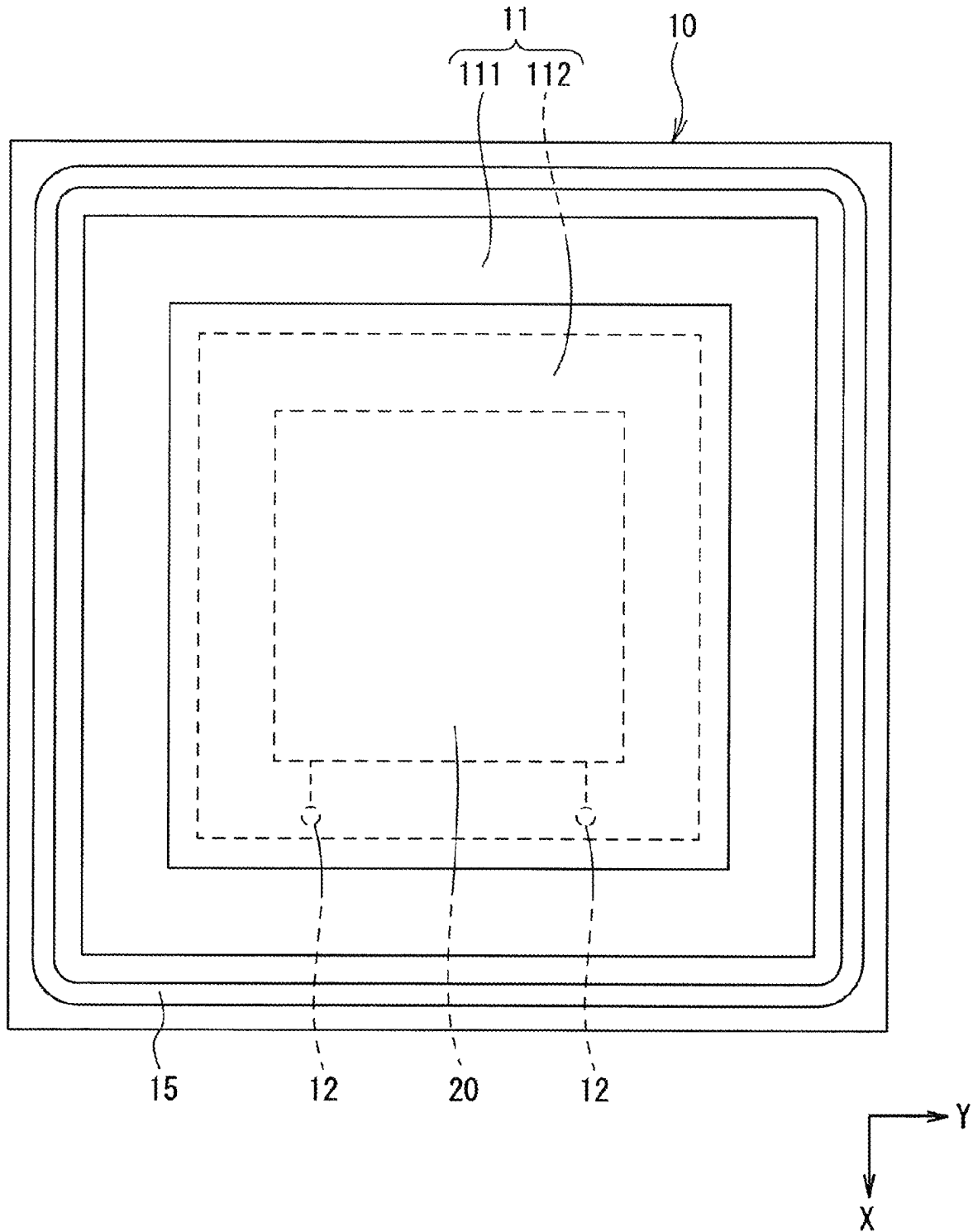


FIG. 6

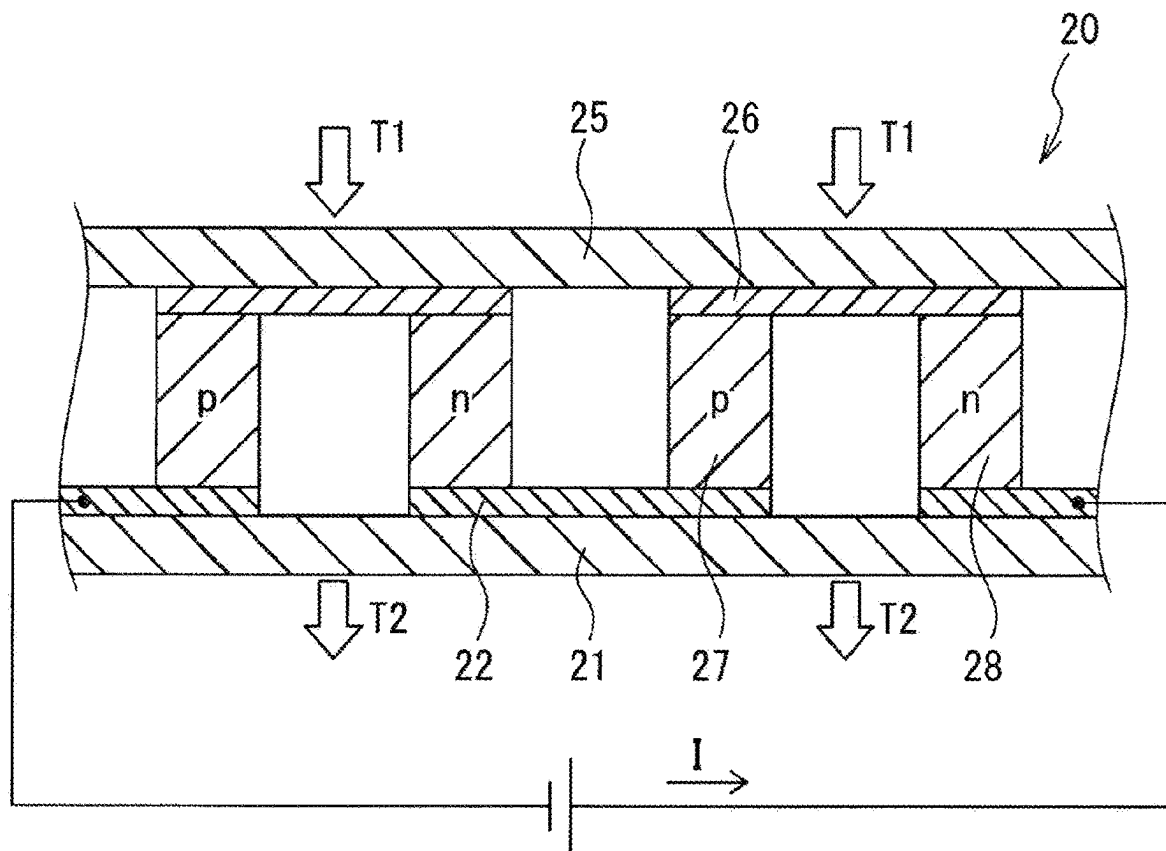


FIG. 7A

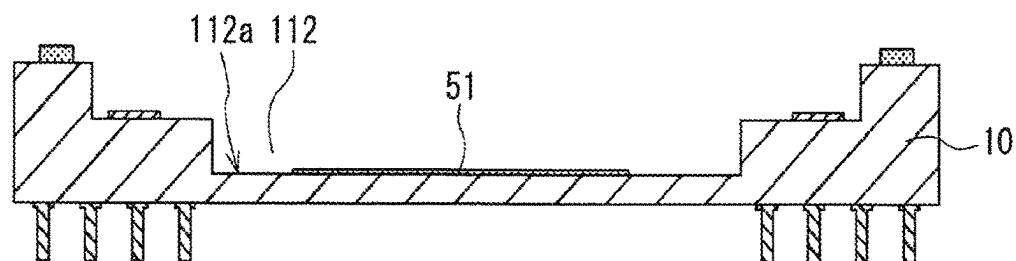


FIG. 7B

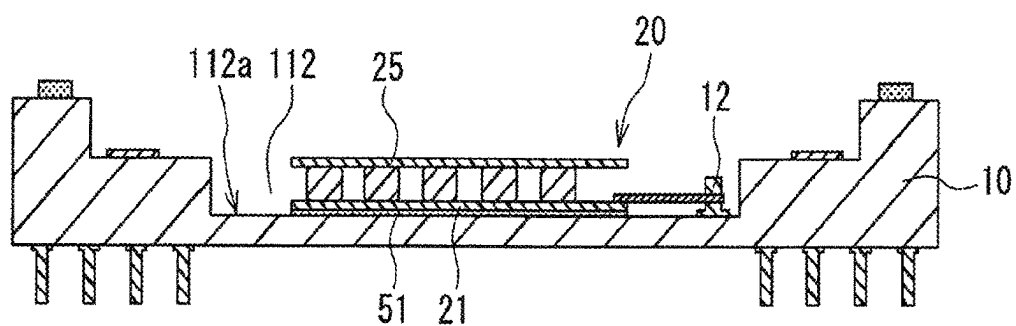


FIG. 7C

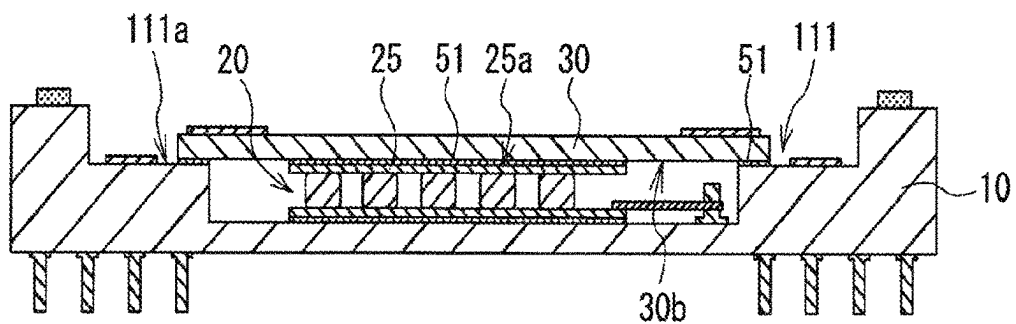
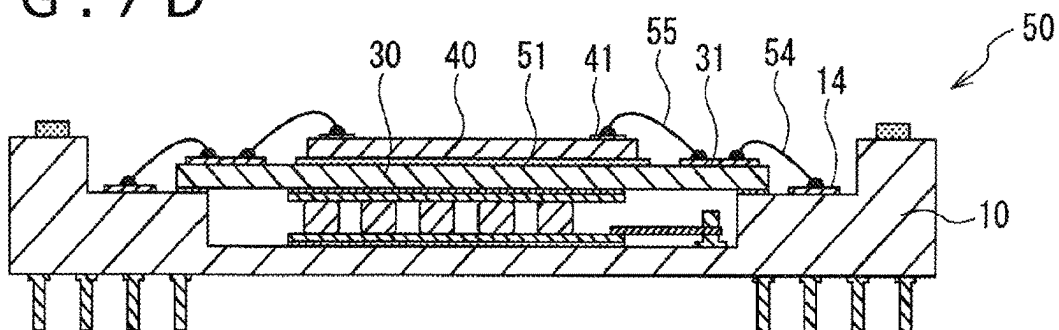


FIG. 7D





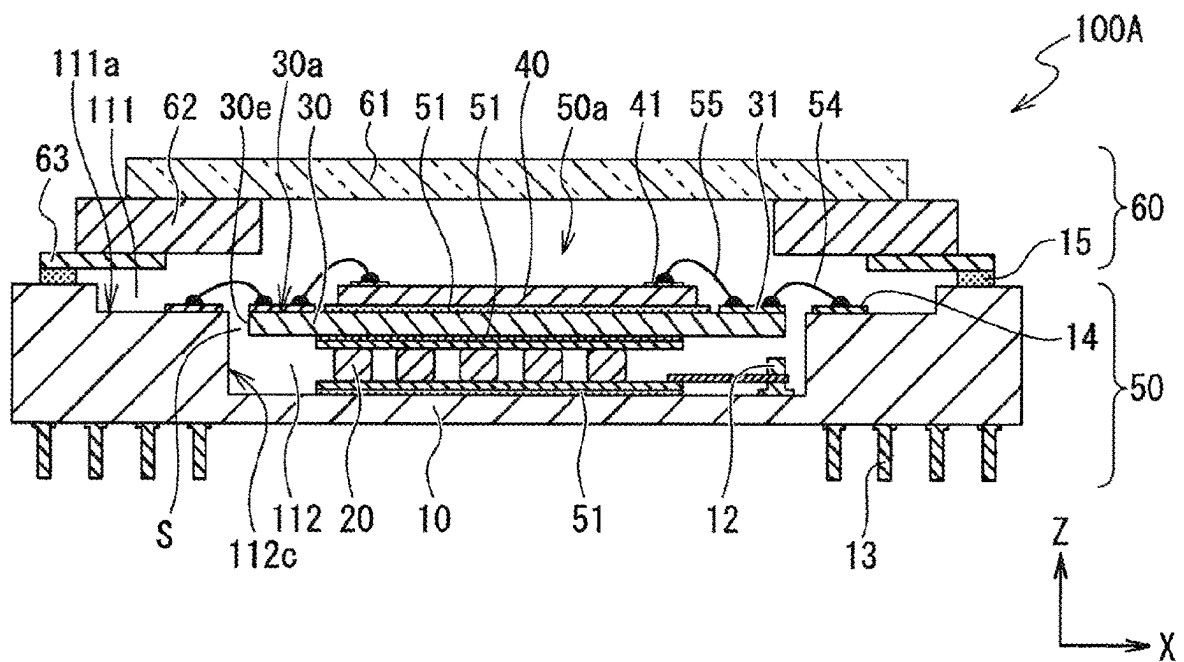


FIG. 9

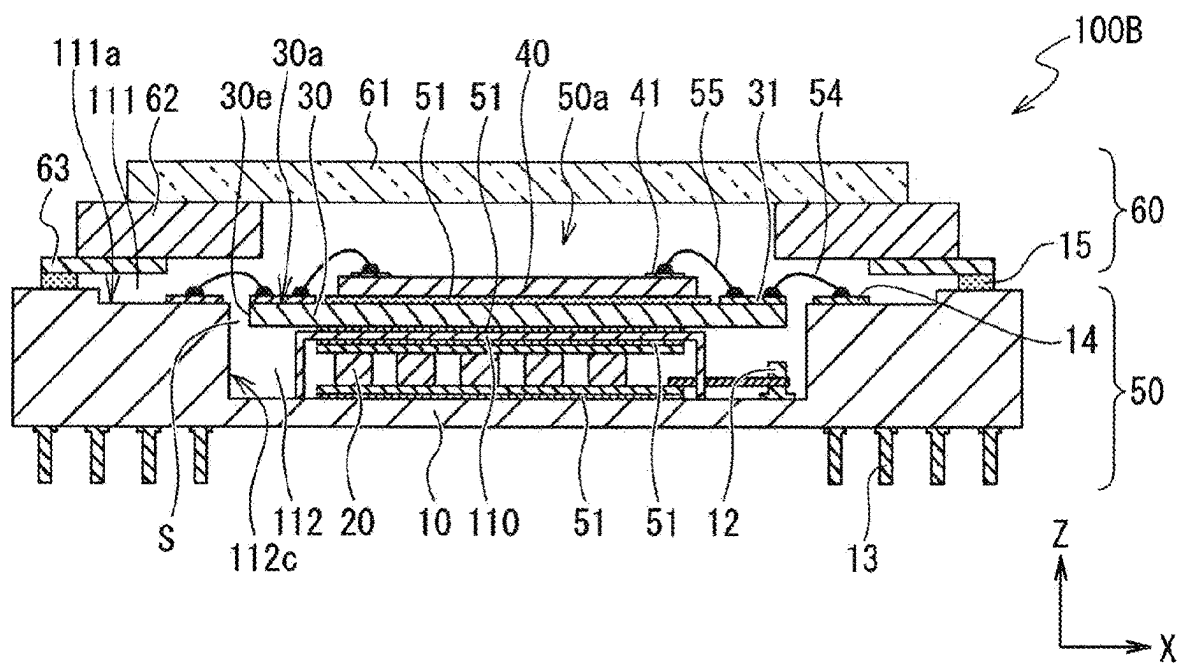


FIG. 10

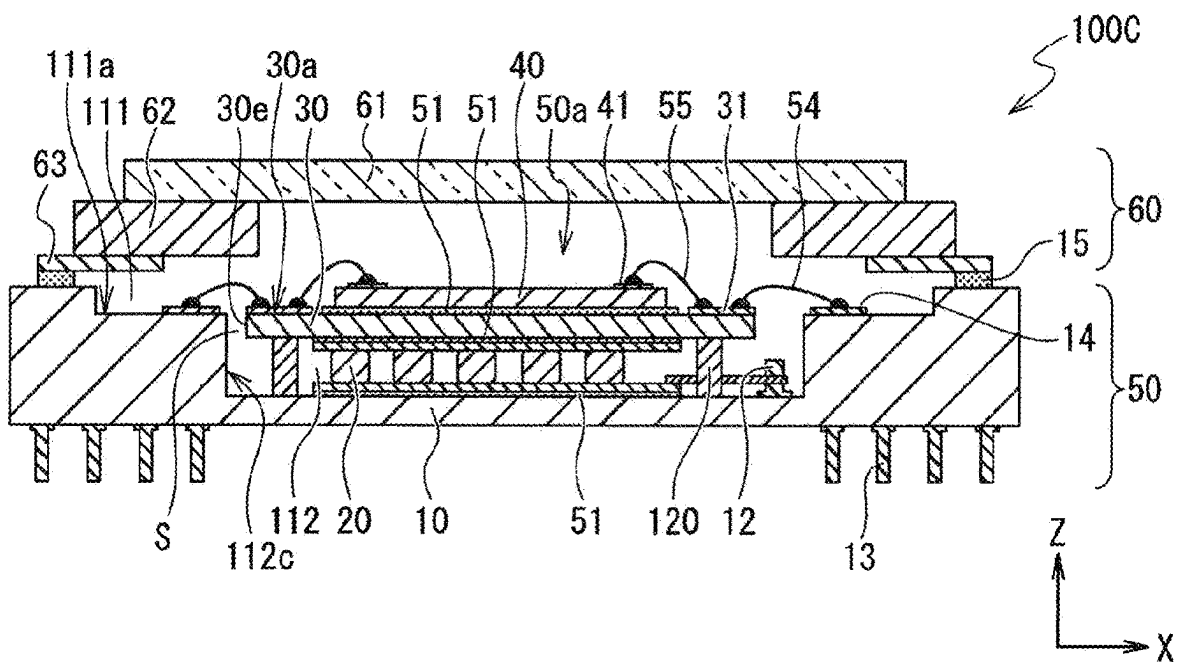


FIG. 11

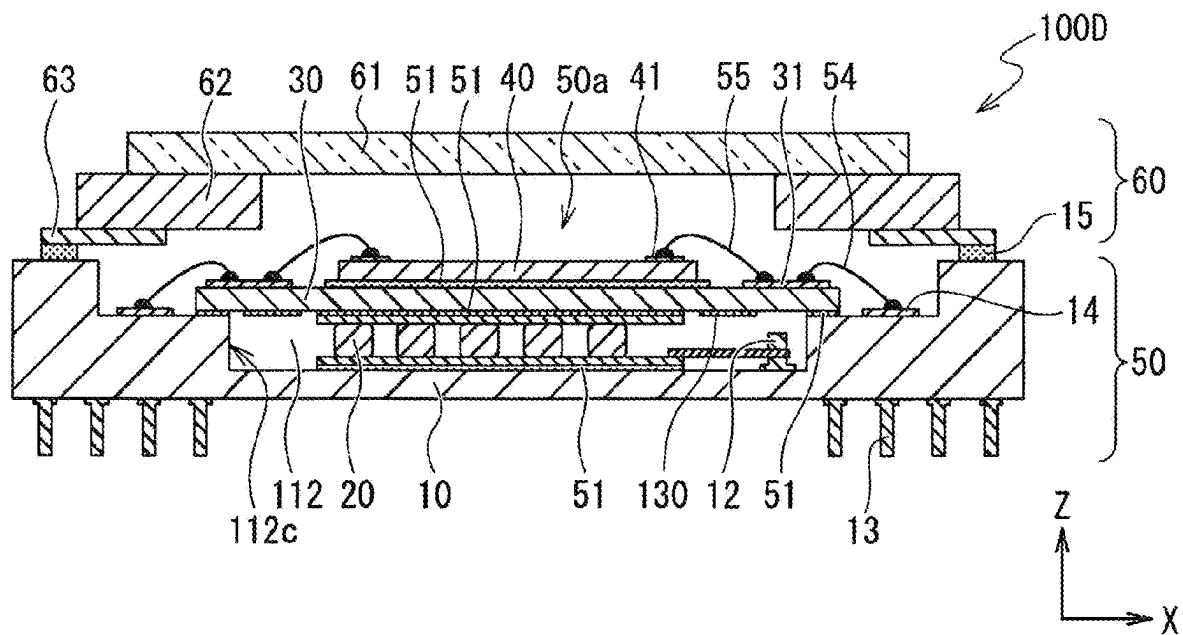
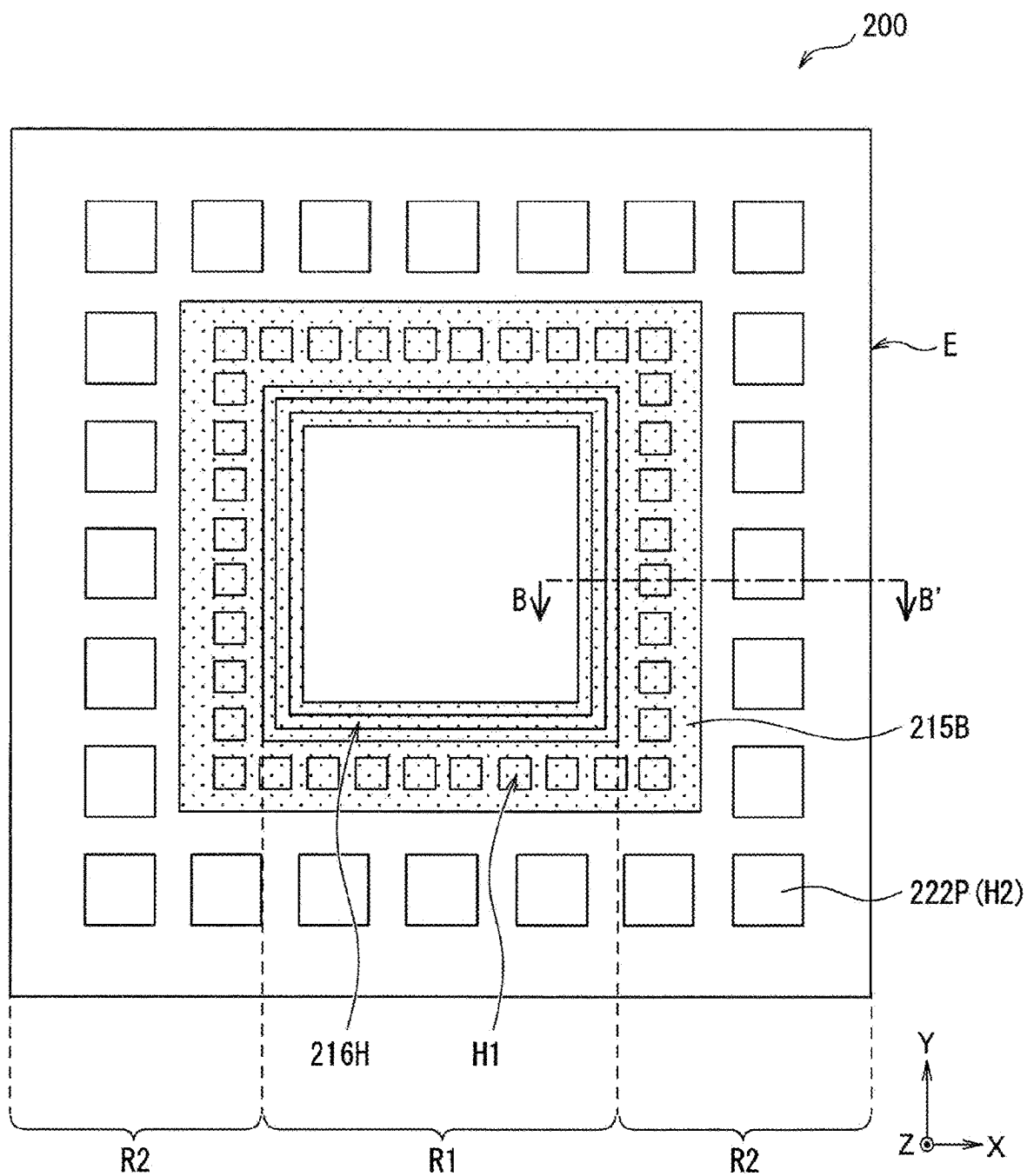


FIG. 12



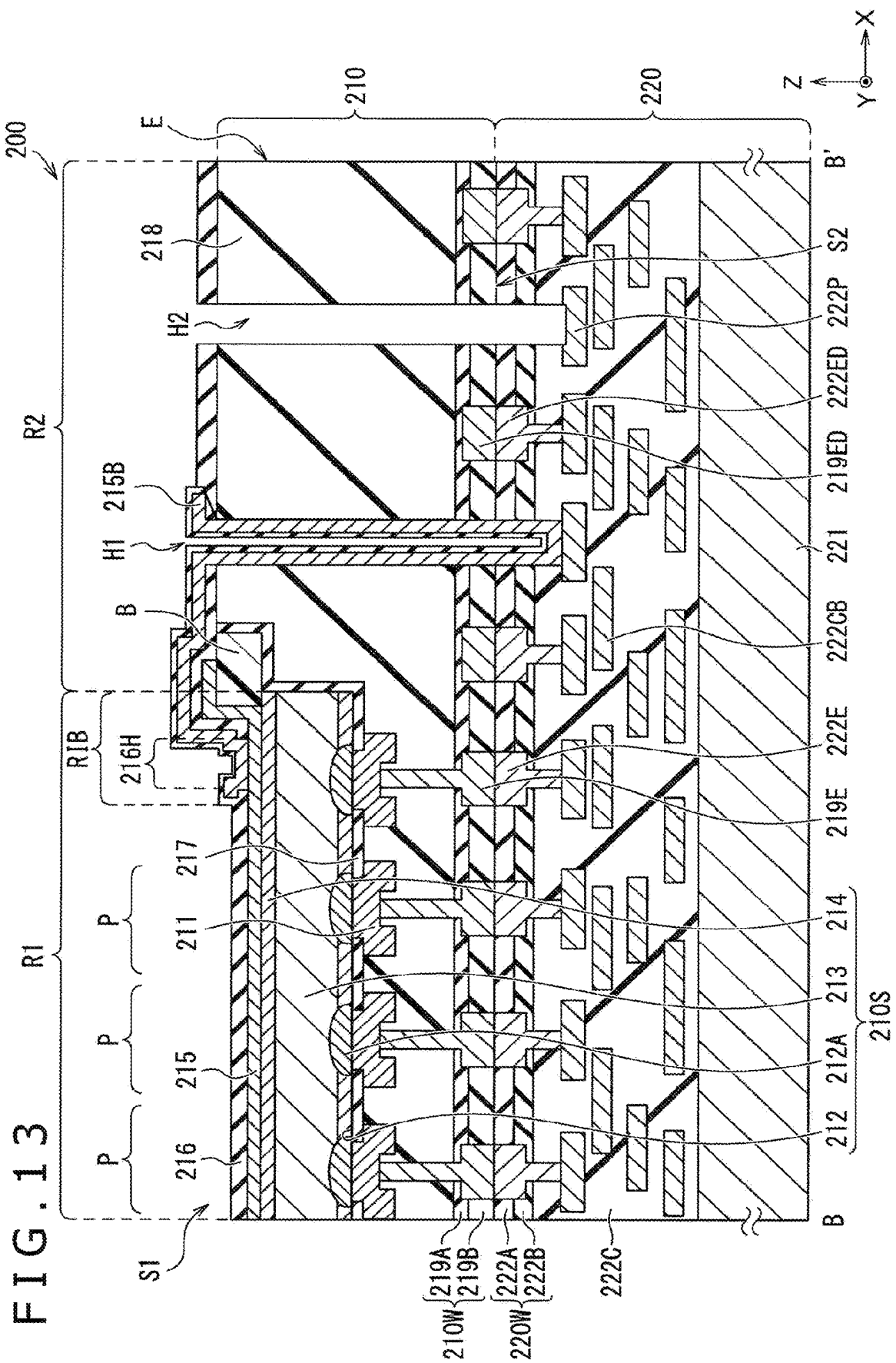
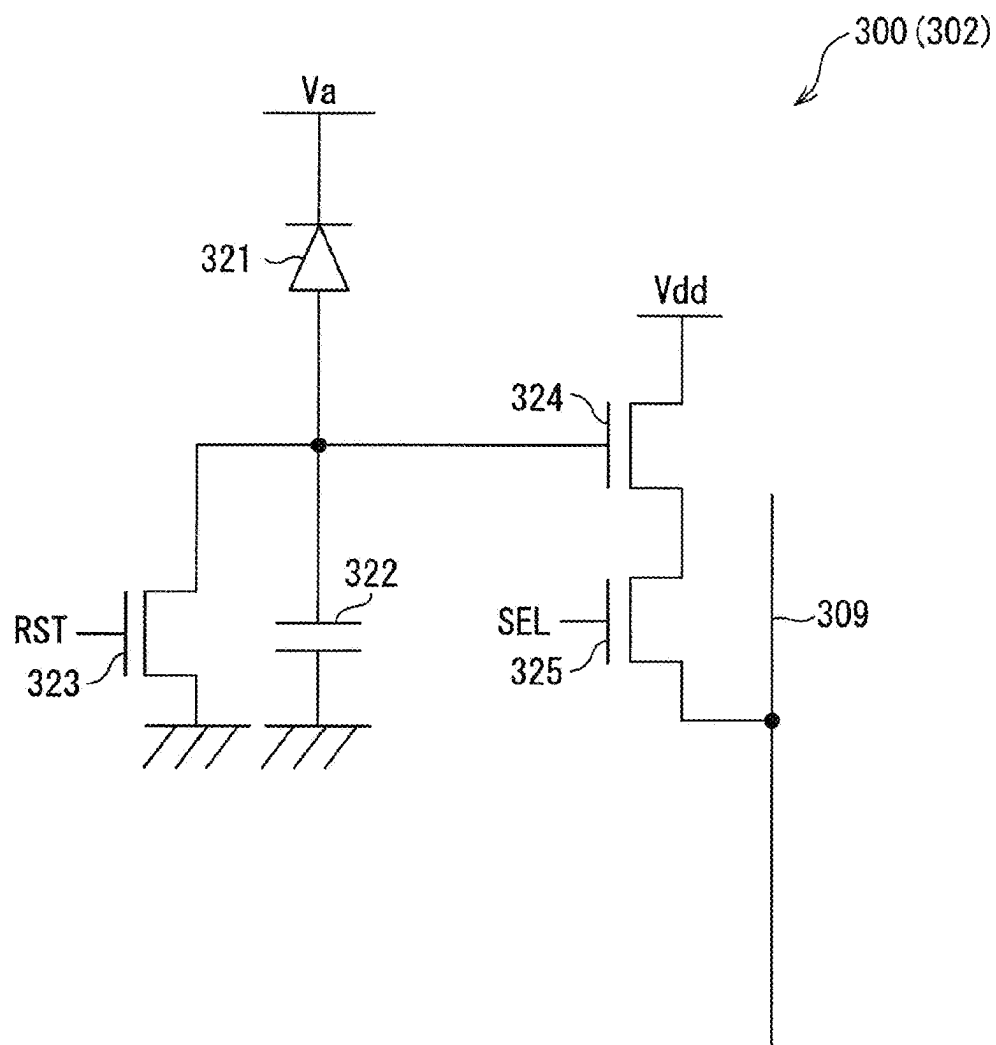
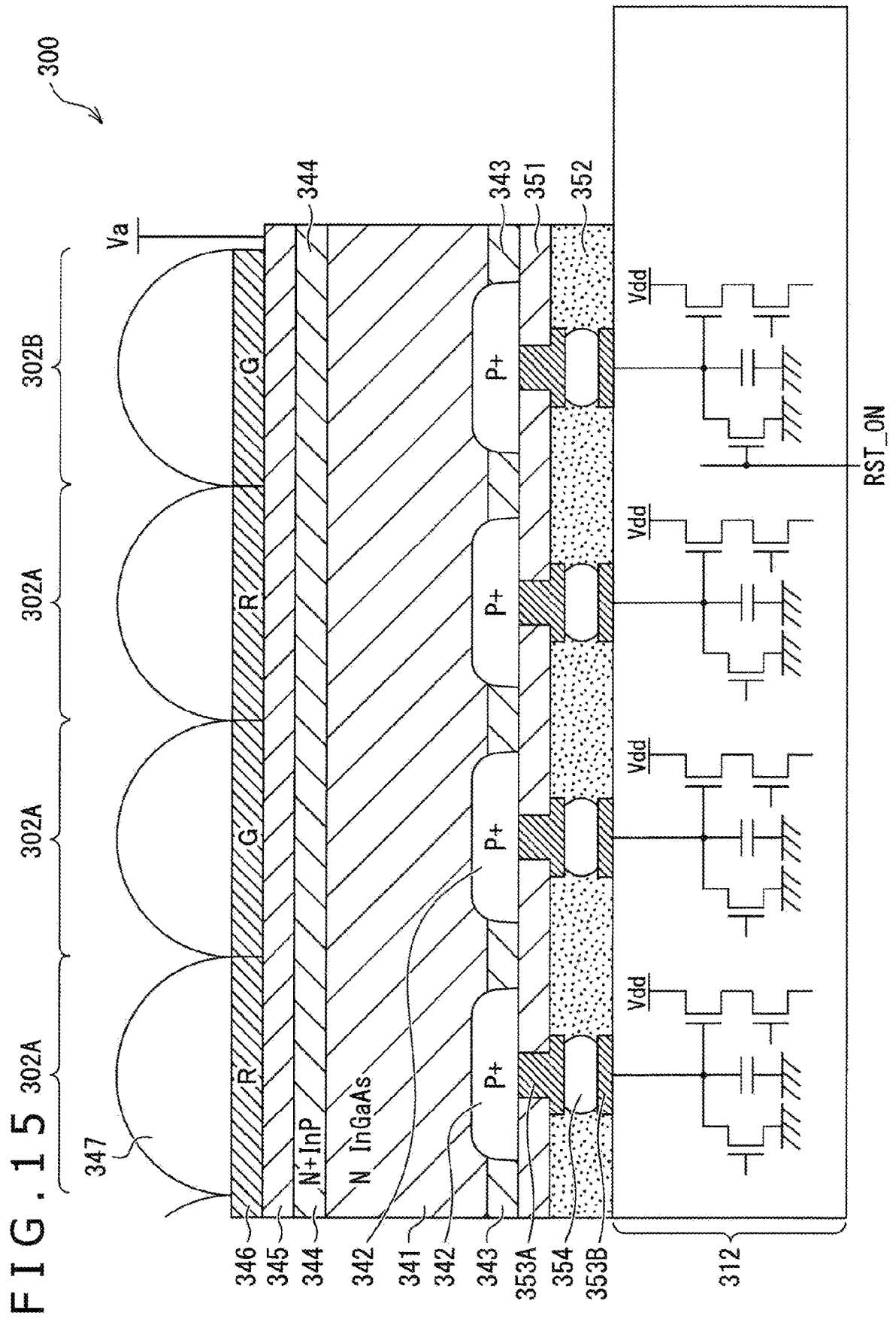


FIG. 14





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## SENSOR APPARATUS

## CROSS REFERENCE TO RELATED APPLICATIONS

This application is a U.S. National Phase of International Patent Application No. PCT/JP2020/004709 filed Feb. 7, 2020, which claims priority benefit of Japanese Patent Application No. JP 2019-068608 filed in the Japan Patent Office on Mar. 29, 2019. Each of the above-referenced applications is hereby incorporated herein by reference in its entirety.

## TECHNICAL FIELD

The present disclosure relates to a sensor apparatus.

## BACKGROUND ART

An airtight sealed package incorporating a Peltier element is known as means of cooling a solid-state imaging apparatus (refer, for example, to PTL 1).

## CITATION LIST

## Patent Literature

[PTL 1]

Japanese Patent Laid-Open No. 2003-258221

## SUMMARY

## Technical Problem

In various processes involving handling of a Peltier element such as a process for manufacturing a Peltier element, a process for transferring the Peltier element, and a process for attaching the Peltier element to a package by using solder or other means, there is a possibility that foreign matters such as dust present in a work environment may adhere inside the Peltier element or to its outer surface. There is a possibility that these foreign matters may detach themselves from the Peltier element, move inside the package, and adhere to a sensor element inside the package, after the Peltier element is arranged inside the package. If a foreign matter adheres to the sensor element, there is a possibility that the foreign matter that has adhered to the sensor element may affect operation of the sensor element.

The present disclosure has been devised in light of the foregoing, and it is an object of the present disclosure to provide a sensor apparatus that can suppress the movement of the foreign matter from the Peltier element to the sensor element.

## Solution to Problem

A mode of the present disclosure is a sensor apparatus that includes a package substrate having a recess portion on a side of a first surface and plural terminals on a side of a second surface located on an opposite side of the first surface, a Peltier element arranged in the recess portion, a circuit substrate arranged on an opposite side of a bottom surface of the recess portion with the Peltier element sandwiched therebetween, and a sensor element attached to an opposite side of a surface of the circuit substrate, the surface being opposed to the Peltier element. According to this, the circuit substrate lies between the Peltier element and the

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sensor element, and an inner side surface of the recess portion is present beside the Peltier element, thus suppressing the movement of foreign matters from the Peltier element to the sensor element.

## BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a plan view illustrating a configuration example of a sensor apparatus according to embodiment 1 of the present disclosure.

FIG. 2 is a cross-sectional view illustrating the configuration example of the sensor apparatus according to embodiment 1 of the present disclosure.

FIG. 3 is an exploded cross-sectional view illustrating the configuration example of the sensor apparatus according to embodiment 1 of the present disclosure.

FIG. 4 is a plan view illustrating a configuration example of an upper surface side of a package substrate according to embodiment 1 of the present disclosure.

FIG. 5 is a plan view illustrating a positional relation between the package substrate and a ceramic interposer substrate according to embodiment 1 of the present disclosure.

FIG. 6 is a cross-sectional view illustrating a configuration example of a Peltier element according to embodiment 1 of the present disclosure.

FIG. 7A is a cross-sectional view illustrating a manufacturing method of the sensor apparatus according to embodiment 1 of the present disclosure.

FIG. 7B is a cross-sectional view illustrating the manufacturing method of the sensor apparatus according to embodiment 1 of the present disclosure.

FIG. 7C is a cross-sectional view illustrating the manufacturing method of the sensor apparatus according to embodiment 1 of the present disclosure.

FIG. 7D is a cross-sectional view illustrating the manufacturing method of the sensor apparatus according to embodiment 1 of the present disclosure.

FIG. 7E is a cross-sectional view illustrating the manufacturing method of the sensor apparatus according to embodiment 1 of the present disclosure.

FIG. 8 is a cross-sectional view illustrating a configuration example of a sensor apparatus according to embodiment 2 of the present disclosure.

FIG. 9 is a cross-sectional view illustrating a configuration example of a sensor apparatus according to embodiment 3 of the present disclosure.

FIG. 10 is a cross-sectional view illustrating a configuration example of a sensor apparatus according to embodiment 4 of the present disclosure.

FIG. 11 is a cross-sectional view illustrating a configuration example of a sensor apparatus according to embodiment 5 of the present disclosure.

FIG. 12 is a plan view illustrating a configuration example of a sensor element according to embodiment 6 of the present disclosure.

FIG. 13 is a cross-sectional view illustrating the configuration example of the sensor element according to embodiment 6 of the present disclosure.

FIG. 14 is a circuit diagram illustrating a pixel circuit of each pixel of a sensor element according to embodiment 7 of the present disclosure.

FIG. 15 is a cross-sectional view illustrating a pixel structure of the sensor element according to embodiment 7 of the present disclosure.

## DESCRIPTION OF EMBODIMENTS

A description will be given below of embodiments of the present disclosure with reference to drawings. In the repre-



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sensation of the drawings referred to in the following description, the same or similar portions will be denoted by the same or similar reference signs. It should be noted, however, that the drawings are schematic and that a relation between thicknesses and planar dimensions, a ratio between the thicknesses of respective layers, and the like are different from the real ones. Accordingly, the following description should be considered to determine the specific thicknesses and dimensions. Also, it is a matter of course that the drawings include the portions where the relation between the dimensions and the ratio differ from one drawing to another.

Also, a definition of directions such as upward and downward directions in the following description is merely for convenience of description and does not restrict a technical idea of the present disclosure. For example, it is a matter of course that if one observes a target by rotating it by 90 degrees, upward and downward directions are changed to leftward and rightward directions for interpretation and that if one observes the target by rotating it by 180 degrees, upward and downward directions are reversed for interpretation.

Also, in the following description, there are cases where directions are described by using wordings "X-axis direction," "Y-axis direction," and "Z-axis direction." For example, the Z-axis direction is the direction along the thickness of a sensor apparatus 100 which will be described later. The X-axis direction and the Y-axis direction are the directions orthogonal to the Z-axis direction. The X-axis direction, the Y-axis direction, and the Z-axis direction are orthogonal to one another. Also, in the following description, the term "plan view" refers to viewing from the Z-axis direction.

#### Embodiment 1

FIG. 1 is a plan view illustrating a configuration example of the sensor apparatus 100 according to embodiment 1 of the present disclosure. FIG. 2 is a cross-sectional view illustrating the configuration example of the sensor apparatus 100 according to embodiment 1 of the present disclosure. FIG. 2 illustrates a cross-section obtained by cutting FIG. 1 along an X-Z plane that passes through an II-II' line.

As illustrated in FIGS. 1 and 2, the sensor apparatus 100 includes a sensor assembly package 50 and a seal-glass-equipped lid 60 (example of a lid) attached to the side of an upper surface 50a of the sensor assembly package 50. The sensor assembly package 50 includes a package substrate 10, a Peltier element 20, a ceramic interposer substrate 30 (example of a circuit substrate), and a sensor element 40. A description will be given first of a configuration of the sensor assembly package 50.

FIG. 3 is an exploded cross-sectional view illustrating the configuration example of the sensor apparatus 100 according to embodiment 1 of the present disclosure. The package substrate 10 is a multi-layer substrate that includes ceramic such as alumina (aluminum oxide) and is, for example, a PGA (Pin Grid Array) substrate. As illustrated in FIG. 3, the package substrate 10 has a first surface (e.g., upper surface 10a) and a second surface (e.g., lower surface 10b) on the opposite side of the first surface. In the package substrate 10, plural wirings are provided therein in multiple layers between the upper surface 10a and the lower surface 10b. These wirings are connected to plural terminals (e.g., pin-shaped terminals 13) provided on the lower surface 10b of the package substrate 10.

FIG. 4 is a plan view illustrating a configuration example of the side of the upper surface 10a of the package substrate

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10 according to embodiment 1 of the present disclosure. As illustrated in FIGS. 3 and 4, a cavity 11 is provided on the side of the upper surface 10a of the package substrate 10. The cavity 11 has a first recess portion 111 and a second recess portion 112 provided on a bottom surface 111a of the first recess portion 111 (examples of a recess portion). The plan-view shape of the first recess portion 111 and the second recess portion 112 is, for example, rectangular. The first recess portion 111 has an opening surface that is larger in diameter than that of the second recess portion 112.

The Peltier element 20 is arranged in the second recess portion 112 and is attached to a bottom surface 112a of the second recess portion 112, for example, via an adhesive 51 (refer to FIG. 2). An upper surface of the Peltier element 20 arranged in the second recess portion 112 (e.g., upper surface 25a of a second ceramic substrate 25 which will be described later) is at the same height or approximately at the same height as that of the bottom surface 111a of the first recess portion 111.

Pin-shaped terminals 12 for connection to lead wires of the Peltier element 20 are provided on the bottom surface 112a of the second recess portion 112. The two pin-shaped terminals 12 are provided. One of the two pin-shaped terminals 12 is connected to a positive lead wire of the Peltier element 20, and the other is connected to a negative lead wire of the Peltier element 20.

As illustrated in FIGS. 2 to 4, a seal ring 15 is provided on the side of the upper surface 10a of a perimeter portion of the package substrate 10. The seal ring 15 is continuously provided in such a manner as to surround the cavity 11 of the package substrate 10 in plan view. The seal ring 15 is a part that is joined to a metallic portion 63, which will be described later, of the seal-glass-equipped lid 60. The seal ring 15 is, for example, iron (Fe)-nickel (Ni)-cobalt (Co) alloy (what is called Kovar) and is surface-treated with Ni and gold (Au) plating or the like.

FIG. 5 is a plan view illustrating a positional relation between the package substrate 10 and the ceramic interposer substrate 30 according to embodiment 1 of the present disclosure. As illustrated in FIG. 2, the side of a lower surface 30b of the ceramic interposer substrate 30 is attached to the bottom surface 111a of the first recess portion 111 and the Peltier element 20 via the adhesive 51. As illustrated in FIGS. 2 and 5, the ceramic interposer substrate 30 is arranged in such a manner as to entirely cover an opening surface of the second recess portion 112.

Plural bonding pads 14 are provided in a region exposed from under the ceramic interposer substrate 30, the region being the bottom surface 111a of the first recess portion 111. Also, plural bonding pads 31 are provided on the side of the lower surface 30b of the ceramic interposer substrate 30. At least some of the plural bonding pads 31 are connected to the bonding pads 14 via wires 54. Also, at least some of the plural bonding pads 31 are connected to bonding pads 41 of the sensor element 40 via wires 55. Alternatively, both the wires 54 and 55 may be connected to the single bonding pad 31. The wires 54 and 55 are, for example, gold wires.

The sensor element 40 is, for example, a CMOS (Complementary Metal Oxide Semiconductor) image sensor or a CCD (Charge Coupled Device) image sensor. A photoelectric conversion layer of the CMOS image sensor or the CCD (Charge Coupled Device) image sensor includes, for example, silicon (Si). Light detected by the sensor element 40 is not limited to visible light and may be, for example, infrared rays or ultraviolet rays. The bonding pads 41 are provided in a perimeter region on the side of the upper surface 40a of the sensor element 40. The side of a lower

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surface 40b of the sensor element 40 is attached to the side of the lower surface 30b of the ceramic interposer substrate 30 via the adhesive 51.

FIG. 6 is a cross-sectional view illustrating a configuration example of the Peltier element 20 according to embodiment 1 of the present disclosure. As illustrated in FIG. 6, the Peltier element 20 has a first ceramic substrate 21, a first copper electrode 22, a second ceramic substrate 25, a second copper electrode 26, a P-type thermoelectric semiconductor 27, and an N-type thermoelectric semiconductor 28. The first copper electrode 22 is provided on the first ceramic substrate 21. The second ceramic substrate 25 is opposed to the first ceramic substrate 21. The second copper electrode 26 is provided on the second ceramic substrate 25. The P-type thermoelectric semiconductor 27 and the N-type thermoelectric semiconductor 28 are each arranged between the first ceramic substrate 21 and the second ceramic substrate 25. The P-type thermoelectric semiconductor 27 and the N-type thermoelectric semiconductor 28 each have its one end connected to the first copper electrode 22 and its other end connected to the second copper electrode 26. The P-type thermoelectric semiconductor 27 and the N-type thermoelectric semiconductor 28 are connected alternately and in series via the first copper electrode 22 and the second copper electrode 26.

As illustrated in FIG. 6, in the Peltier element 20, when a direct current is passed from the N-type thermoelectric semiconductor 28, the second ceramic substrate 25 absorbs heat T1 (heat absorption), and the first ceramic substrate 21 radiates heat T2 (heat radiation). The second ceramic substrate 25 is attached to the ceramic interposer substrate 30 via the adhesive 51, and the first ceramic substrate 21 is attached to the package substrate 10 via the adhesive 51. This makes it possible for the Peltier element 20 to dissipate heat generated by the sensor element 40 or the like from the ceramic interposer substrate 30 to the package substrate 10.

A description will be given next of a configuration of the sensor assembly package 50. As illustrated in FIGS. 1 to 3, the seal-glass-equipped lid 60 has a seal glass 61, a ceramic frame 62, and the metallic portion 63. The ceramic frame 62 is provided on the side of a lower surface 61b of a perimeter portion of the seal glass 61. The metallic portion 63 is provided on the side of a lower surface 62b of the ceramic frame 62. The seal glass 61 and the ceramic frame 62 are joined together, for example, with low melting point glass. The ceramic frame 62 and the metallic portion 63 are joined together, for example, with an Ag—Cu wax material or the like.

The metallic portion 63 is a part joined to the seal ring 15 of the package substrate 10, for example, by seam welding or other means. The metallic portion 63 includes the same material as that of the seal ring 15, one example of which is iron (Fe)-nickel (Ni)-cobalt (Co) alloy (what is called Kovar) and is surface-treated with Ni and gold (Au) plating or the like. The seal-glass-equipped lid 60 is joined to the side of the upper surface 50a of the sensor assembly package 50, and the side of the upper surface 50a of the sensor assembly package 50 is sealed airtight.

A description will be given next of a manufacturing method of the sensor apparatus 100. It should be noted that various apparatuses such as an application apparatus for applying the adhesive 51, an attachment apparatus for attaching the Peltier element 20, the ceramic interposer substrate 30, or the seal-glass-equipped lid 60, a wire bonding apparatus, and a seam welding apparatus are used to manufacture the sensor apparatus 100. In the embodiments of the present disclosure, these apparatuses will be

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collectively referred to as manufacturing apparatuses. Also, at least some of tasks performed by the manufacturing apparatuses may be performed by workers.

FIGS. 7A, 7B, 7C, 7D, and 7E are cross-sectional views illustrating the manufacturing method of the sensor apparatus 100 according to embodiment 1 of the present disclosure. First, as illustrated in FIG. 7A, the manufacturing apparatus applies the adhesive 51 to the bottom surface 112a of the second recess portion 112 of the package substrate 10. Next, as illustrated in FIG. 7B, the manufacturing apparatus attaches the Peltier element 20 to the bottom surface 112a of the second recess portion 112 via the adhesive 51. Also, before or after this bonding, the manufacturing apparatus wraps the lead wires of the Peltier element 20 around the pin-shaped terminals 12 and solders the lead wires. This allows the Peltier element 20 to be electrically connected to the package substrate 10.

Next, as illustrated in FIG. 7C, the manufacturing apparatus attaches the ceramic interposer substrate 30 to the Peltier element 20 and the package substrate 10. For example, the manufacturing apparatus applies the adhesive 51 to the bottom surface 111a of the first recess portion 111 and to the upper surface 25a of the second ceramic substrate 25 which is at the same height as that of the bottom surface 111a. Then, the manufacturing apparatus attaches the lower surface 30b of the ceramic interposer substrate 30 to the bottom surface 111a of the first recess portion 111 and to the upper surface 25a of the second ceramic substrate 25 via the adhesive 51. This allows the opening surface of the second recess portion 112 to be entirely covered with the ceramic interposer substrate 30. A path that runs through a space between the Peltier element 20 and the sensor element 40 is closed by the ceramic interposer substrate 30 and an inner side surface 112c of the second recess portion 112, thereby separating the space around the Peltier element 20 from the space around the sensor element 40.

Next, as illustrated in FIG. 7D, the manufacturing apparatus attaches the sensor element 40 to an upper surface 30a of the ceramic interposer substrate 30 via the adhesive 51. Then, the manufacturing apparatus electrically connects the bonding pads 41 of the sensor element 40 and the bonding pads 31 of the ceramic interposer substrate 30 via the wires 55. Also, the manufacturing apparatus electrically connects the bonding pads 31 of the ceramic interposer substrate 30 and the bonding pads 14 of the package substrate 10 via the wires 55. This completes the sensor assembly package 50.

Next, as illustrated in FIG. 7E, with the seal-glass-equipped lid 60 and the sensor assembly package 50 aligned with each other, the manufacturing apparatus attaches the seal-glass-equipped lid 60 to the sensor assembly package 50, for example, by seam welding or other means. This allows the space between the seal-glass-equipped lid 60 and the sensor assembly package (hereinafter referred to as a "package interior") to be sealed airtight.

Seam welding is a type of resistance welding and is a continuous welding method that uses a roller electrode and achieves welding by rotating the electrode while applying a pressure and passing a current. The seal ring 15 of the sensor assembly package 50 and the metallic portion 63 of the seal-glass-equipped lid 60 are joined by this method. By placing the inside of a chamber provided in the seam welding apparatus into a dry air, nitrogen, vacuum, or other atmosphere, this process can also keep the atmosphere inside the package in a dry air, nitrogen, or vacuum atmosphere.

As described above, the sensor apparatus 100 according to embodiment 1 of the present disclosure includes the

package substrate 10, the Peltier element 20, the ceramic interposer substrate 30, and the sensor element 40. The package substrate 10 has the second recess portion 112 on the side of the upper surface 10a and has the plural pin-shaped terminals 13 on the side of the lower surface 10b located on the opposite side of the upper surface 10a. The Peltier element 20 is arranged in the second recess portion 112. The ceramic interposer substrate 30 is arranged on the opposite side of the bottom surface 112a of the second recess portion 112 with the Peltier element 20 sandwiched therebetween. In the ceramic interposer substrate 30, the sensor element 40 is attached to the opposite side of the surface opposed to the Peltier element 20.

According to this, the ceramic interposer substrate 30 lies between the Peltier element 20 and the sensor element 40, and the inner side surface 112c of the second recess portion 112 is present beside the Peltier element 20. Also, there is no gap S between the inner side surface 112c of the second recess portion 112 and the ceramic interposer substrate 30. The space around the Peltier element 20 is separated from the space around the sensor element 40 by the inner side surface 112c of the second recess portion 112 and the ceramic interposer substrate 30, thus preventing the movement of foreign matters from the Peltier element 20 to the sensor element 40. This makes it possible, even in a case where foreign matters such as dust adhere inside the Peltier element 20 or to its outer surface, to reduce the number of foreign matters that adhere to the sensor element 40 and to decrease the possibility that the operation of the sensor element 40 is affected due to foreign matters.

Also, the sensor apparatus 100 further includes the seal-glass-equipped lid 60 that is attached to the package substrate 10 and covers the ceramic interposer substrate 30 and the sensor element 40. The space between the seal-glass-equipped lid 60 and the package substrate 10 is sealed airtight. According to this, it is possible to prevent intrusion of foreign matters from outside into the sensor apparatus 100. It is possible to further reduce the number of foreign matters that adhere to the sensor element 40.

Also, the seal-glass-equipped lid 60 is attached to the package substrate 10 by seam welding. According to this, it is possible to improve reliability of airtight seal inside the package.

Also, the package substrate 10 is a multi-layer substrate that includes ceramic. The seal-glass-equipped lid 60 includes the seal glass 61, the ceramic frame 62, and the metallic portion 63. Neither of the package substrate 10 and the seal-glass-equipped lid 60 included in the package of the sensor apparatus 100 includes resin. This makes it possible to prevent the intrusion of moisture and the like from outside into the sensor apparatus 100 with high reliability.

#### Embodiment 2

In above embodiment 1, it has been described that the ceramic interposer substrate 30 is arranged in such a manner as to entirely cover the opening surface of the second recess portion 112 and that the space around the Peltier element 20 is separated from the space around the sensor element 40 by the ceramic interposer substrate 30 and the inner side surface 112c of the second recess portion 112. However, the embodiments of the present disclosure are not limited thereto. For example, the ceramic interposer substrate 30 may be arranged in such a manner as to partially cover the opening surface of the second recess portion 112. The Peltier element 20 and the sensor element 40 may not be completely spatially separated.

FIG. 8 is a cross-sectional view illustrating a configuration example of a sensor apparatus 100A according to embodiment 2 of the present disclosure. As illustrated in FIG. 8, in the sensor apparatus 100A, the ceramic interposer substrate 30 partially, and not completely, covers the opening surface of the second recess portion 112. A plan-view size of the ceramic interposer substrate 30 is approximately equal to or slightly smaller than that of the opening surface of the second recess portion 112. Because of this structure, the gap S is present between the inner side surface 112c of the second recess portion 112 and an edge portion 30e of the ceramic interposer substrate 30. There is a possibility that foreign matters may pass through the gap S.

However, the ceramic interposer substrate 30 protrudes more outward than the Peltier element 20, and the path that runs through the space between the Peltier element 20 and the sensor element 40 detours the edge portion 30e of the ceramic interposer substrate 30. This lengthens a distance of the path that runs through the above space. Also, the inner side surface 112c of the second recess portion 112 is present beside the Peltier element 20. The inner side surface 112c of the second recess portion 112 is in proximity to the edge portion 30e of the ceramic interposer substrate 30, and the gap S is narrower.

As described above, the path that runs through the above space has a long distance, and further, the gap S is narrower, thus suppressing the movement of foreign matters from the Peltier element 20 to the sensor element 40. This makes it possible, even in the case where foreign matters such as dust adhere inside the Peltier element 20 or to its outer surface, to reduce the number of foreign matters that adhere to the sensor element 40 and to decrease the possibility that the operation of the sensor element 40 is affected due to foreign matters.

Also, it is preferred that, in the sensor apparatus 100A, the upper surface 30a of the ceramic interposer substrate 30 should be at the same or approximately at the same height as that of the bottom surface 111a of the first recess portion 111, as illustrated in FIG. 8. This can not only reduce the above gap S but also eliminate a difference in level between the bonding pads 14 and 31, thus making it possible to shorten the length of the wires 54 that electrically connect the bonding pads 14 and 31. As a result, it becomes possible to reduce an amount of Au used that is included in the wires 54, thus contributing to reduced manufacturing cost of the sensor apparatus 100A.

#### Embodiment 3

In above embodiment 1, it has been described that the space around the Peltier element 20 is separated from the space around the sensor element 40 by the ceramic interposer substrate 30 and the inner side surface 112c of the second recess portion 112. In the embodiments of the present disclosure, however, the structure for separating the space between the Peltier element 20 and the sensor element 40 is not limited to that described above.

FIG. 9 is a cross-sectional view illustrating a configuration example of a sensor apparatus 100B according to embodiment 3 of the present disclosure. As illustrated in FIG. 9, the sensor apparatus 100B includes a metallic casing 110 that is arranged in the second recess portion 112 and that accommodates the Peltier element 20 therein. The casing 110 is, for example, a cuboid and shaped in such a manner that the side of a surface opposed to the bottom surface 112a of the second recess portion 112 is open. The inside of the casing 110 is fastened to the Peltier element 20 via the

adhesive 51, and the outside of the casing 110 is fastened to the ceramic interposer substrate 30 via the adhesive 51. The space around the Peltier element 20 is separated from the space around the sensor element 40 by the casing 110.

In such a configuration, the casing 110 prevents the movement of foreign matters from the Peltier element 20 to the sensor element 40. This makes it possible, even in the case where foreign matters such as dust adhere inside the Peltier element 20 or to its outer surface, to reduce the number of foreign matters that adhere to the sensor element 40 and to decrease the possibility that the operation of the sensor element 40 is affected due to foreign matters.

Also, the casing 110 is metallic and covers the Peltier element 20. This allows the casing 110 to shield the Peltier element 20, thus suppressing electromagnetic waves (noise) generated by the Peltier element 20 from entering the sensor element 40.

A description will be given of a process for attaching the casing 110. For example, after the Peltier element 20 is attached to the bottom surface 112a of the second recess portion 112 and the lead wires of the Peltier element 20 are electrically connected to the pin-shaped terminals 12, the manufacturing apparatus applies the adhesive 51 to the upper surface of the Peltier element 20 and places the casing 110 over the Peltier element 20 to which the adhesive 51 has been applied. This allows the casing 110 to be fastened to the upper surface of the Peltier element 20 via the adhesive 51. Thereafter, the manufacturing apparatus applies the adhesive 51 to an upper surface of the casing 110 and attaches the ceramic interposer substrate 30 to the upper surface of the casing 110 to which the adhesive 51 has been applied. The subsequent processes are the same as in embodiment 1.

#### Embodiment 4

In the embodiments of the present disclosure, the structure for separating the space between the Peltier element 20 and the sensor element 40 may be realized by a component other than the casing.

FIG. 10 is a cross-sectional view illustrating a configuration example of a sensor apparatus 100C according to embodiment 4 of the present disclosure. As illustrated in FIG. 10, the sensor apparatus 100C includes a wall member 120 that is sandwiched between the bottom surface 112a of the second recess portion 112 and the ceramic interposer substrate 30 and laterally surrounds the Peltier element 20. For example, the wall member 120 includes resin. The plan-view shape of the wall member 120 is a frame, and the Peltier element 20 is arranged inside the frame. The space around the Peltier element 20 is separated from the space around the sensor element 40 by the ceramic interposer substrate 30 and the wall member 120.

In such a configuration, the ceramic interposer substrate 30 and the wall member 120 prevent the movement of foreign matters from the Peltier element 20 to the sensor element 40. This makes it possible, even in the case where foreign matters such as dust adhere inside the Peltier element 20 or to its outer surface, to reduce the number of foreign matters that adhere to the sensor element 40 and to decrease the possibility that the operation of the sensor element 40 is affected due to foreign matters.

It should be noted that, in the embodiments of the present disclosure, the material included in the wall member 120 is not limited to resin and may be a metal. In a case where the wall member 120 includes a metal, the wall member 120 can

laterally shield the Peltier element 20, thus reducing noise that propagates horizontally (e.g., direction parallel to the X-Y plane).

A description will be given of a process for forming the wall member 120. For example, after the Peltier element 20 is attached to the bottom surface 112a of the second recess portion 112 and the lead wires of the Peltier element 20 are electrically connected to the pin-shaped terminals 12, the manufacturing apparatus forms the wall member 120 on the bottom surface 112a in such a manner as to surround the periphery of the Peltier element 20. Although the wall member 120 may be formed in any manner, an example of forming method is to attach the wall member 120, which is formed in advance by resin molding, to the bottom surface 112a via the adhesive (not illustrated). Thereafter, the manufacturing apparatus attaches the ceramic interposer substrate 30 to the upper surface of the Peltier element 20 via the adhesive 51. The adhesive 51 may or may not be present between the wall member 120 and the ceramic interposer substrate 30. The subsequent processes are the same as in embodiment 1.

#### Embodiment 5

In the embodiments of the present disclosure, the structure for reducing noise may be provided on the ceramic interposer substrate 30.

FIG. 11 is a cross-sectional view illustrating a configuration example of a sensor apparatus 100D according to embodiment 5 of the present disclosure. As illustrated in FIG. 11, in the sensor apparatus 100D, a magnetic shield pattern 130 (example of a conductor layer) is provided on the side of the lower surface 30b of the ceramic interposer substrate 30. The magnetic shield pattern 130 is provided on the side of the lower surface 30b of the ceramic interposer substrate 30 in such a manner as to surround a region opposed to the Peltier element 20 on the lower surface 30b of the ceramic interposer substrate 30. The magnetic shield pattern 130 is fixed at a given potential (e.g., ground voltage: 0V). In such a configuration, the magnetic shield pattern 130 can shield the Peltier element 20, thus suppressing noise generated by the Peltier element 20 from entering the sensor element 40.

#### Embodiment 6

In above embodiments 1 to 5, the sensor element 40 has been illustrated as an example of a "sensor element" of the present disclosure, and it has been described that the sensor element 40 is a CMOS image sensor or a CCD image sensor. However, the "sensor element" of the present disclosure is not limited to a CMOS image sensor or a CCD image sensor. Also, the material included in the photoelectric conversion layer of the "sensor element" of the present disclosure is not limited to silicon (Si) and may be a compound semiconductor.

For example, the "sensor element" of the present disclosure may be an InGaAs sensor whose photoelectric conversion layer includes a group III-V semiconductor such as InGaAs (indium-gallium-arsenide) and may be, for example, a sensor element 200 illustrated in FIGS. 12 and 13. Even in such a mode, it is possible to suppress the movement of foreign matters from the Peltier element 20 to the sensor element 200. A description will be given below of a configuration example of the sensor element 200.

FIGS. 12 and 13 are a plan view and a cross-sectional view illustrating a configuration example of the sensor

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element **200** according to embodiment 6 of the present disclosure. FIG. **12** illustrates a planar configuration of the sensor element **200**, and FIG. **13** illustrates a cross-sectional configuration thereof along line B-B' in FIG. **12**. The sensor element **200** is applied to, for example, an infrared sensor using a compound semiconductor material such as a group III-V semiconductor and has a function to photoelectrically convert light of a wavelength ranging from a visible region (e.g., 380 nm or more to less than 780 nm) to a short-infrared region (e.g., 780 nm or more to less than 2400 nm). In the sensor element **200**, plural light reception unit regions p (pixels P) that are, for example, arranged two-dimensionally are provided (FIG. **13**).

The sensor element **200** has an element region R1 in the center and a peripheral region R2 that is provided in the outside of the element region R1 and that surrounds the element region R1 (FIG. **12**). The sensor element **200** has a conductive film **215B** that is provided to span from the element region R1 to the peripheral region R2. The conductive film **215B** has an opening in a region opposed to a center portion of the element region R1.

The sensor element **200** has a layered structure in which an element substrate **210** and a readout circuit substrate **220** are stacked (FIG. **13**). One surface of the element substrate **210** is a light entry surface (light entry surface S1), and the surface on the opposite side of the light entry surface S1 (other surface) is a junction surface with the readout circuit substrate **220** (junction surface S2).

The element substrate **210** has a wiring layer **210W**, first electrodes **211**, a semiconductor layer **210S** (first semiconductor layer), a second electrode **215**, and a passivation film **216** in sequence from the position close to the readout circuit substrate **220**. A surface of the semiconductor layer **210S**, the surface being opposed to the wiring layer **210W**, and an edge surface (side surface) thereof are covered with an insulating film **217**. The readout circuit substrate **220** is what is called a ROIC (Readout integrated circuit) and has a wiring layer **220W**, a multi-layer wiring layer **222C**, and a semiconductor substrate **221**. The wiring layer **220W** is in contact with the junction surface S2 of the element substrate **210**. The semiconductor substrate **221** is opposed to the element substrate **210** with the wiring layer **220W** and the multi-layer wiring layer **222C** therebetween.

The element substrate **210** has the semiconductor layer **210S** in the element region R1. In other words, a region where the semiconductor layer **210S** is provided is the element region R1 of the sensor element **200**. Of the element region R1, a region exposed from the conductive film **215B** (region opposed to the opening of the conductive film **215B**) is a light reception region. Of the element region R1, a region covered with the conductive film **215B** is an OPB (Optical Black) region R1B. The OPB region R1B is provided in such a manner as to surround the light reception region. The OPB region R1B is used to acquire a black level pixel signal.

The element substrate **210** has a filling-in layer **218** together with the insulating film **217** in the peripheral region R2. Holes H1 and H2 that penetrate the element substrate **210** and reach the readout circuit substrate **220** are provided in the peripheral region R2. In the sensor element **200**, light enters the semiconductor layer **210S** from the light entry surface S1 of the element substrate **210** via the passivation film **216**, the second electrode **215**, and a second contact layer **214**. A signal charge acquired as a result of photoelectric conversion by the semiconductor layer **210S** moves via the first electrodes **211** and the wiring layer **210W** and is read

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out by the readout circuit substrate **220**. A description will be given below of the configuration of each portion.

The wiring layer **210W** is provided to span from the element region R1 to the peripheral region R2 and has the junction surface S2 with the readout circuit substrate **220**. In the sensor element **200**, the junction surface S2 of the element substrate **210** is provided in the element region R1 and the peripheral region R2, and, for example, the junction surface S2 of the element region R1 and the junction surface S2 of the peripheral region R2 form the same plane. In the sensor element **200**, the junction surface S2 of the peripheral region R2 is formed as a result of the provision of the filling-in layer **218**.

The wiring layer **210W** has contact electrodes **219E** and dummy electrodes **219ED**, for example, inside interlayer insulating films **219A** and **219B**. For example, the interlayer insulating film **219B** is provided on the side of the readout circuit substrate **220**, and an interlayer insulating film **19A** is provided on the side of a first contact layer **212**, and the interlayer insulating films **219A** and **219B** are stacked one on top of another. The interlayer insulating films **219A** and **219B** each include, for example, an inorganic insulating material. Silicon nitride (SiN), aluminum oxide (Al<sub>2</sub>O<sub>3</sub>), silicon oxide (SiO<sub>2</sub>), hafnium oxide (HfO<sub>2</sub>), and the like are examples of this inorganic insulating material. The interlayer insulating films **219A** and **219B** may include the same inorganic insulating material.

The contact electrodes **219E** are provided, for example, in the element region R1. The contact electrodes **219E** are used to electrically connect the first electrodes **211** and the readout circuit substrate **220** and are each provided for each pixel P in the element region R1. The contact electrodes **219E** adjacent to each other are electrically separated from each other by the filling-in layer **218** and the interlayer insulating films **219A** and **219B**. The contact electrodes **219E** each include, for example, a copper (Cu) pad and are exposed on the junction surface S2. The dummy electrodes **219ED** are provided, for example, in the peripheral region R2. The dummy electrodes **219ED** are connected to dummy electrodes **222ED** of the wiring layer **220W**. By providing the dummy electrodes **219ED** and the dummy electrodes **222ED**, it is possible to improve strength of the peripheral region R2. The dummy electrodes **219ED** are formed, for example, by the same process as that of the contact electrodes **219E**. The dummy electrodes **219ED** each include, for example, a copper (Cu) pad and are exposed on the junction surface S2.

The first electrodes **211** provided between the contact electrodes **219E** and the semiconductor layer **210S** are electrodes (anodes) to which a voltage for reading out a signal charge (holes or electrons; a description will be given hereinafter assuming that the signal charge is holes for reasons of convenience) generated by a photoelectric conversion layer **213** and are each provided for each pixel P in the element region R1. The first electrodes **211** are provided in such a manner as to fill in the openings of the insulating film **217** and are in contact with the semiconductor layer **210S** (more specifically, a diffusion region **212A**). The first electrodes **211** are, for example, larger than the openings of the insulating film **217**, and part of the first electrodes **211** is provided in the filling-in layer **218**. That is, an upper surface of the first electrodes **211** (the surface on the side of the semiconductor layer **210S**) is in contact with the diffusion region **212A**, and part of a lower surface and a side surface of the first electrodes **211** is in contact with the filling-in layer **218**. The first electrodes **211** adjacent to each other are

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electrically separated from each other by the insulating film **217** and the filling-in layer **218**.

The first electrodes **211** include, for example, any one of titanium (Ti), tungsten (W), titanium nitride (TiN), platinum (Pt), gold (Au), germanium (Ge), palladium (Pd), zinc (Zn), nickel (Ni), and aluminum (Al) or an alloy that includes at least one of them. The first electrodes **211** may be a single film of one of these materials or a layered film that includes a combination of two or more of these materials. For example, the first electrodes **211** include a layered film of titanium and tungsten. The first electrodes **211** are, for example, several tens to hundreds of nanometers in thickness.

The semiconductor layer **210S** includes, for example, the first contact layer **212**, the photoelectric conversion layer **213**, and the second contact layer **214** from the position close to the wiring layer **210W**. The first contact layer **212**, the photoelectric conversion layer **213**, and the second contact layer **214** have the same planar shape, and edge surfaces of the respective layers are arranged at the same position in plan view.

The first contact layer **212** is provided, for example, in common for all the pixels P and arranged between the insulating film **217** and the photoelectric conversion layer **213**. The first contact layer **212** is used to electrically separate the pixels P adjacent to each other, and the plural diffusion regions **212A**, for example, are provided in the first contact layer **212**. By using, to form the first contact layer **212**, a compound semiconductor material having a bandgap larger than that of the compound semiconductor material included in the photoelectric conversion layer **213**, it is also possible to suppress dark current. For example, N-type InP (indium phosphide) can be used for the first contact layer **212**.

The diffusion regions **212A** provided in the first contact layer **212** are provided apart from each other. The diffusion regions **212A** are each provided for each pixel P, and the first electrode **211** is connected to each of the diffusion regions **212A**. The diffusion region **212A** is also provided in the OPB region R1B. The diffusion regions **212A** are used to read out the signal charge generated by the photoelectric conversion layer **213** for each pixel P and each include, for example, a P-type impurity. Zn (zinc) or the like is an example of a P-type impurity. As described above, a pn junction interface is formed between the diffusion region **212A** and the first contact layer **212** other than the diffusion region **212A**, to electrically separate the pixels P adjacent to each other. The diffusion regions **212A** are provided, for example, in the direction of the thickness of the first contact layer **212** and also provided in part of the direction of the thickness of the photoelectric conversion layer **213**.

The photoelectric conversion layer **213** between the first electrodes **211** and the second electrode **215**, and more specifically, the photoelectric conversion layer **213** between the first contact layer **212** and the second contact layer **214**, is provided, for example, in common for all the pixels P. The photoelectric conversion layer **213** generates a signal charge by absorbing light having a specific wavelength and includes, for example, a compound semiconductor material such as an i-type group III-V semiconductor. InGaAs (indium-gallium-arsenic), InAsSb (indium-arsenic-antimony), InAs (indium-arsenic), InSb (indium-antimony), HgCdTe (mercury-cadmium-tellurium), and the like are examples of a compound semiconductor material included in the photoelectric conversion layer **213**. The photoelectric conversion layer **213** may include Ge (germanium). The photoelectric conversion layer **213** photoelectrically converts, for

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example, light of a wavelength ranging from the visible region to the short-infrared region.

The second contact layer **214** is provided, for example, in common for all the pixels P. The second contact layer **214** is provided between the photoelectric conversion layer **213** and the second electrode **215** and is in contact with them. The second contact layer **214** is a region where the charge emitted by the second electrode **215** moves and includes, for example, a compound semiconductor including an N-type impurity. N-type InP (indium phosphide), for example, can be used for the second contact layer **214**.

The second electrode **215** is provided on the second contact layer **214** (light entry side) in such a manner as to be in contact with the second contact layer **214**, for example, as a common electrode for each pixel P. The second electrode **215** is used to emit, of the charge generated by the photoelectric conversion layer **213**, the charge that is not used as a signal charge (cathode). For example, in a case where holes are read out from the first electrodes **211** as a signal charge, electrons, for example, can be emitted through the second electrode **215**. The second electrode **215** includes, for example, a conductive film that allows passage of incident light such as infrared rays. ITO (Indium Tin Oxide), ITiO ( $\text{In}_2\text{O}_3\text{—TiO}_2$ ), or the like can be used, for example, for the second electrode **215**. The second electrode **215** may be provided, for example, in a lattice pattern to partition the pixels P adjacent to each other. A conductive material having low light transmissivity can be used for the second electrode **215**.

The passivation film **216** covers the second electrode **215** from the side of the light entry surface S1. The passivation film **216** may have an anti-reflection function. For example, silicon nitride (SiN), aluminum oxide ( $\text{Al}_2\text{O}_3$ ), silicon oxide ( $\text{SiO}_2$ ), tantalum oxide ( $\text{Ta}_2\text{O}_5$ ), and the like can be used for the passivation film **216**. The passivation film **216** has an opening **216H** in the OPB region R1B. The opening **216H** is provided in the shape of a picture frame surrounding the light reception region (FIG. 12). The opening **216H** may be, for example, a rectangular or circular hole in plan view. The conductive film **215B** is electrically connected to the second electrode **215** by the opening **216H** of the passivation film **216**.

The insulating film **217** is provided between the first contact layer **212** and the filling-in layer **218**, covers the edge surface of the first contact layer **212**, the edge surface of the photoelectric conversion layer **213**, the edge surface of the second contact layer **214**, and the edge surface of the second electrode **215**, and is in contact with the passivation film **216** in the peripheral region R2. The insulating film **217** includes oxide such as silicon oxide ( $\text{SiO}_x$ ) or aluminum oxide ( $\text{Al}_2\text{O}_3$ ). The insulating film **217** may have a layered structure that includes plural films. The insulating film **217** may include a silicon-based insulating film such as silicon oxide nitride (SiON), silicon oxide including carbon (SiOC), silicon nitride (SiN), and silicon carbide (SiC). The insulating film **217** is, for example, several tens to hundreds of nanometers in thickness.

The conductive film **215B** is provided to span from the OPB region R1B to a hole H1 in the peripheral region R2. The conductive film **215B** is in contact with the second electrode **215** in the opening **216H** of the passivation film **216** provided in the OPB region R1B and is in contact with a wiring **222CB** of the readout circuit substrate **220** via the hole H1. This allows a voltage to be supplied to the second electrode **215** from the readout circuit substrate **220** via the conductive film **215B**. The conductive film **215B** not only functions as a voltage supply path to the second electrode

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**215** as described above but also has a function as a light-shielding film and forms the OPB region **R1B**. The conductive film **215B** includes, for example, a metallic material that includes tungsten (W) aluminum (Al), titanium (Ti), molybdenum (Mo), tantalum (Ta), or copper (Cu). A passivation film may be provided on the conductive film **215B**.

An adhesive layer **B** may be provided between the edge portion of the second contact layer **214** and the second electrode **215**. The adhesive layer **B** is used to form the sensor element **200** and plays a role of joining the semiconductor layer **210S** to a temporary substrate. The adhesive layer **B** includes, for example, tetraethoxysilane (TEOS), silicon oxide (SiO<sub>2</sub>), or the like. The adhesive layer **B** is provided, for example, wider than the edge surface of the semiconductor layer **210S** and is covered with the filling-in layer **218** together with the semiconductor layer **210S**. The insulating film **217** is provided between the adhesive layer **B** and the filling-in layer **218**.

#### Embodiment 7

The “sensor element” of the present disclosure may be a sensor element **300** illustrated in FIGS. **14** and **15**. Even in such a mode, it is possible to suppress the movement of foreign matters from the Peltier element **20** (refer to FIG. **2**) to the sensor element **300**. A description will be given below of a configuration example of the sensor element **300**.

FIG. **14** is a circuit diagram illustrating a pixel circuit of each pixel **302** of the sensor element **300** according to embodiment 7 of the present disclosure. FIG. **15** is a cross-sectional view illustrating a pixel structure of the sensor element **300** according to embodiment 7 of the present disclosure. A description will be given first of the pixel circuit illustrated in FIG. **14**.

As illustrated in FIG. **14**, each pixel **302** includes a photoelectric conversion section **321**, a capacitive element **322**, a reset transistor **323**, an amplifying transistor **324**, and a selection transistor **325**. The photoelectric conversion section **321** includes a semiconductor thin film using a compound semiconductor such as InGaAs and generates a charge (signal charge) proportional to an amount of light received. A given bias voltage  $V_a$  is applied to the photoelectric conversion section **321**.

The capacitive element **322** stores the charge generated by the photoelectric conversion section **321**. The capacitive element **322** can include, for example, at least one of P-N junction capacitance, MOS capacitance, or wiring capacitance. When the reset transistor **323** is turned ON by a reset signal **RST**, the charge stored in the capacitive element **322** is emitted to a source (ground), thus resetting a potential of the capacitive element **322**.

The amplifying transistor **324** outputs a pixel signal proportional to the stored potential of the capacitive element **322**. That is, the amplifying transistor **324** forms a source follower circuit with a load MOS (not illustrated) as a constant current source connected via a vertical signal line **309**. The pixel signal indicating a level proportional to the charge stored in the capacitive element **322** is output from the amplifying transistor **324** to a signal processing circuit (not illustrated) via the selection transistor **325**.

A description will be given next of the pixel structure illustrated in FIG. **15**. It should be noted that each pixel **302** in a pixel array region can be classified into a normal pixel **302A** and a charge ejection pixel **302B**, depending on the difference in the manner of controlling the reset transistor **323**. It should be noted, however, that the normal pixel **302A** and the charge ejection pixel **302B** have the same pixel

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structure. Accordingly, these pixels will be simply described as the pixels **302**. It should be noted that the charge ejection pixel **302B** is arranged at an outermost side of the pixel array region.

The readout circuit for the photoelectric conversion section **321**, the capacitive element **322**, the reset transistor **323**, the amplifying transistor **324**, and the selection transistor **325** of each pixel **302** illustrated in FIG. **14** is formed for each pixel in a semiconductor substrate **312** that includes a monocrystalline material such as monocrystalline silicon (Si). It should be noted that reference signs such as the capacitive element **322**, the reset transistor **323**, the amplifying transistor **324**, and the selection transistor **325** illustrated in FIG. **14** are omitted in FIG. **15**.

As illustrated in FIG. **15**, an N-type semiconductor thin film **341** that will be the photoelectric conversion section **321** (refer to FIG. **14**) is formed over the entire surface of the pixel array region on the upper side of the semiconductor substrate **312** which is the light entry side. InGaP, InAlP, InGaAs, and InAlAs may be used as the N-type semiconductor thin film **341**, and a compound semiconductor having a chalcopyrite structure may also be used. The compound semiconductor having a chalcopyrite structure is a material that offers a high light absorption coefficient and high sensitivity over a wide wavelength range and is preferably used as the N-type semiconductor thin film **341** for photoelectric conversion. Such a compound semiconductor having a chalcopyrite structure includes elements around group IV elements such as Cu, Al, GaIn, S, and Se, and CuGaInS-based mixed crystal, CuAlGaInS-based mixed crystal, CuAlGaInSSe-based mixed crystal, and the like are examples thereof.

Also, in addition to the compound semiconductor described above, amorphous silicon (Si), germanium (Ge), a quantum-dot photoelectric conversion film, an organic photoelectric conversion film, and the like can also be used as the material of the N-type semiconductor thin film **341**. In the present embodiment, it is assumed that an InGaAs compound semiconductor is used as the N-type semiconductor thin film **341**.

A high-concentration P-type layer **342** included in a pixel electrode is formed for each pixel on the lower side of the N-type semiconductor thin film **341**, the lower side being the side of the semiconductor substrate **312**. Then, an N-type layer **343** as a pixel separation region for separating the pixels **302** is formed between the high-concentration P-type layers **342** that are each formed for each pixel. The N-type layers **343** are each formed by using, for example, a compound semiconductor such as InP. The N-type layers **343** play a role of preventing dark current in addition to their function as the pixel separation region.

Meanwhile, a high-concentration N-type layer **344** having higher concentration than the N-type semiconductor thin film **341** is formed on the upper side of the N-type semiconductor thin film **341**, the upper side being the light entry side. The high-concentration N-type layer **344** is formed by using the compound semiconductor such as InP that is used as the pixel separation region. The high-concentration N-type layer **344** functions as a barrier layer for preventing a charge generated by the N-type semiconductor thin film **341** from flowing in reverse. A compound semiconductor such as InGaAs, InP, and InAlAs can be used as the material of the high-concentration N-type layer **344**, for example.

An anti-reflection film **45** is formed on the high-concentration N-type layer **344** being as the barrier layer. Silicon nitride (SiN), hafnium oxide (HfO<sub>2</sub>), aluminum oxide (Al<sub>2</sub>O<sub>3</sub>), zirconium oxide (ZrO<sub>2</sub>), tantalum oxide (Ta<sub>2</sub>O<sub>5</sub>),

titanium oxide (TiO<sub>2</sub>), and the like can be used as the material of an anti-reflection film **345**, for example.

Any one of the high-concentration N-type layer **344** or the anti-reflection film **345** also functions as an upper electrode of the two electrodes that vertically sandwich the N-type semiconductor thin film **341**, and the given voltage V<sub>a</sub> is applied to the high-concentration N-type layer **344** or the anti-reflection film **345** as the upper electrode.

Color filters **346** and an on-chip lens **347** are further formed on the anti-reflection film **345**. The color filters **346** allow passage of any one of R (red), G (green), or B (blue) light (wavelength light) and are arranged, for example, in what is called a Bayer pattern in the pixel array region.

A passivation layer **351** and an insulating layer **352** are formed on the lower side of the N-type layer **343** being as the pixel separation region and of the high-concentration N-type layer **344** included in the pixel electrode. Then, connection electrodes **353A** and **353B** and a bump electrode **354** are formed in such a manner as to penetrate the passivation layer **351** and the insulating layer **352**. The connection electrodes **353A** and **353B** and the bump electrode **354** electrically connect the high-concentration P-type layer **342** included in the pixel electrode and the capacitive element **322** that stores the charge.

#### Other Embodiments

As described above, the present disclosure has been described by using the embodiments and modification examples. However, the statement and the drawings which are part of this disclosure should not be construed as limiting the present disclosure. Various alternative embodiments, working examples, and operational technologies will become apparent to a person skilled in the art from this disclosure. In the embodiments of the present disclosure, for example, a heatsink may be attached to the package substrate **10** to improve heat radiation of the Peltier element **20** to the package substrate **10**. A region that is on the side of the lower surface **10b** of the package substrate **10** and that is opposed to the Peltier element **20** is an example of a region where the heatsink is attached.

As described above, it is a matter of course that the present technology includes various embodiments and the like not described herein. At least one of various omissions, substitutions, and alterations can be made to components without departing from the gist of the embodiments and modification examples described above. Also, advantageous effects described in the present specification are merely illustrative and not restrictive, and there may be additional advantageous effects.

It should be noted that the present disclosure can also have the following configurations.

- (1) A sensor apparatus including:
  - a package substrate having a recess portion on a side of a first surface and plural terminals on a side of a second surface located on an opposite side of the first surface;
  - a Peltier element arranged in the recess portion;
  - a circuit substrate arranged on an opposite side of a bottom surface of the recess portion with the Peltier element sandwiched therebetween; and
  - a sensor element attached to an opposite side of a surface of the circuit substrate, the surface being opposed to the Peltier element.

- (2) The sensor apparatus of feature (1), further including:
  - a lid that is attached to the package substrate and covers the circuit substrate and the sensor element, in which a space between the lid and the package substrate is sealed airtight.
- (3) The sensor apparatus of feature (2), in which the lid is attached to the package substrate by seam welding.
- (4) The sensor apparatus of any one of features (1) to (3), in which
  - a space around the Peltier element is separated from a space around the sensor element by the circuit substrate and an inner side surface of the recess portion.
- (5) The sensor apparatus of any one of features (1) to (3), further including:
  - a casing that is arranged in the recess portion and accommodates the Peltier element, in which
  - a space around the Peltier element is separated from a space around the sensor element by the casing.
- (6) The sensor apparatus of feature (5), in which the casing is metallic.
- (7) The sensor apparatus of any one of features (1) to (3), further including:
  - a wall member that is sandwiched between the bottom surface of the recess portion and the circuit substrate and laterally surrounds the Peltier element, in which
  - a space around the Peltier element is separated from a space around the sensor element by the circuit substrate and the wall member.
- (8) The sensor apparatus of any one of features (1) to (5) and (7), in which
  - the circuit substrate has a conductor layer provided in such a manner as to surround a region opposed to the Peltier element.
- (9) The sensor apparatus of any one of features (1) to (8), in which
  - the sensor element has a photoelectric conversion layer that includes silicon.
- (10) The sensor apparatus of any one of features (1) to (8), in which
  - the sensor element has a photoelectric conversion layer that includes a compound semiconductor.

#### REFERENCE SIGNS LIST

- 10**: Package substrate
- 10a, 25a, 30a, 40a**: Upper surface
- 10b, 30b, 40b, 62b**: Lower surface
- 11**: Cavity
- 12, 13**: Pin-shaped terminal
- 14**: Bonding pad
- 15**: Seal ring
- 20**: Peltier element
- 21**: First ceramic substrate
- 22**: First copper electrode
- 25**: Second ceramic substrate
- 26**: Second copper electrode
- 27**: P-type thermoelectric semiconductor



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28: N-type thermoelectric semiconductor  
 30: Ceramic interposer substrate  
 30e: Edge portion  
 31: Bonding pad  
 40, 200, 300: Sensor element  
 41: Bonding pad  
 50: Sensor assembly package  
 50a: Upper surface  
 51: Adhesive  
 54, 55: Wire  
 60: Seal-glass-equipped lid  
 61b: Lower surface  
 62: Ceramic frame  
 63: Metallic portion  
 100, 100A, 100B, 100C, 100D: Sensor apparatus  
 110: Casing  
 111: First recess portion  
 111a, 112a: Bottom surface  
 112: Second recess portion  
 112c: Inner side surface  
 120: Wall member  
 130: Magnetic shield pattern  
 210: Element substrate  
 210S: Semiconductor layer  
 210W: Wiring layer  
 211: First electrodes  
 212: First contact layer  
 212A: Diffusion region  
 213: Photoelectric conversion layer  
 214: Second contact layer  
 215: Second electrode  
 215B: Conductive film  
 216: Passivation film  
 216H: Opening  
 217: Insulating film  
 218: Filling-in layer  
 219A, 219B: Interlayer insulating film  
 219E: Contact electrode  
 219ED: Dummy electrode  
 220: Readout circuit substrate  
 220W: Wiring layer  
 221: Semiconductor substrate  
 222C: Multi-layer wiring layer  
 222CB: Wiring  
 222ED: Dummy electrode  
 302: Pixels  
 302A: Normal pixel  
 302B: Charge ejection pixel  
 309: Vertical signal line  
 312: Semiconductor substrate  
 321: Photoelectric conversion section  
 322: Capacitive element  
 323: Reset transistor  
 324: Amplifying transistor  
 325: Selection transistor  
 341: Semiconductor thin film  
 342: P-type layer  
 343, 344: N-type layer  
 345: Anti-reflection film  
 346: Color filter  
 347: On-chip lens  
 351: Passivation layer  
 352: Insulating layer  
 353A, 353B: Connection electrode  
 354: Bump electrode

## 20

The invention claimed is:

1. A sensor apparatus, comprising:
  - a package substrate that includes:
    - a first recess portion on a first side of a first surface of the package substrate;
    - a second recess portion on a bottom surface of the first recess portion; and
    - a plurality of terminals on a second side of a second surface of the package substrate, wherein the second side is opposite to the first side;
  - a Peltier element in the second recess portion;
  - a circuit substrate on an opposite side of a bottom surface of the second recess portion, wherein the Peltier element is between a bottom surface of the circuit substrate and the bottom surface of the second recess portion, the bottom surface of the circuit substrate is attached to the bottom surface of the first recess portion, and the bottom surface of the circuit substrate faces the Peltier element; and
  - an image sensor on an upper surface of the circuit substrate, wherein the circuit substrate is different from the image sensor, and the upper surface of the circuit substrate is opposite to the bottom surface of the circuit substrate.
2. The sensor apparatus according to claim 1, further comprising:
  - a lid attached to the package substrate; and
  - a sealed airtight space between the lid and the package substrate, wherein the lid is configured to cover the circuit substrate and the image sensor.
3. The sensor apparatus according to claim 2, wherein the lid is seam welded to the package substrate.
4. The sensor apparatus according to claim 1, wherein the circuit substrate and an inner side surface of the second recess portion separate a space around the Peltier element from a space around the image sensor.
5. The sensor apparatus according to claim 1, further comprising a casing in the second recess portion, wherein the casing accommodates the Peltier element, and the casing separates a space around the Peltier element from a space around the image sensor.
6. The sensor apparatus according to claim 5, wherein the casing includes a metal.
7. The sensor apparatus according to claim 1, further comprising a wall member between the circuit substrate and the bottom surface of the second recess portion, wherein the wall member laterally surrounds the Peltier element, and the circuit substrate and the wall member separate a space around the Peltier element from a space around the image sensor.
8. The sensor apparatus according to claim 1, wherein the bottom surface of the circuit substrate includes a conductor layer, and the conductor layer surrounds a region opposed to the Peltier element.
9. The sensor apparatus according to claim 1, wherein the image sensor has a photoelectric conversion layer that includes silicon.
10. The sensor apparatus according to claim 1, wherein the image sensor has a photoelectric conversion layer that includes a compound semiconductor.

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